2nd Graduate Summer Institute "Complex Plasmas"

# Theory and application of Langmuir probes

**I. Korolov**

Greifswald 2010

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- Basics of probe diagnostic
- Working regimes of Langmuir probe
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# Introduction



Irving Langmuir
1881 – 1957
Nobel Prize 1932

One of the most frequently employed methods for plasma diagnostics is Langmuir probes. They were introduced by Langmuir in the 1920s*.

The advantage of the probe method is that we can measure local parameters of studied plasma almost all other techniques, such as spectroscopy or microwave propagation, give information averaged over a large volume of plasma.

Another advantage is the simplicity of the used equipment that allows us to receive result quickly and without high experience.

The probe method enables the performing of the time- as well as the space-resolved measurements of plasma parameters

Plasma parameters

Plasma density ($n_e$, $n_i$) Temperature ($T_e$) EEDF Ion flux Plasma potential

Working area:

Pressures: $p \sim 10^{-5} - 10^{3}$ Torr

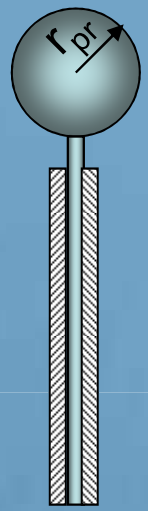
Concentration of electrons $n_e \sim 10^{6} - 10^{15}$ cm$^{-3}$.

*I. Langmuir and H. Mott-Smith, “The theory of collectors in gaseous discharges,” Phys. Rev. **28**, 727–763 1926

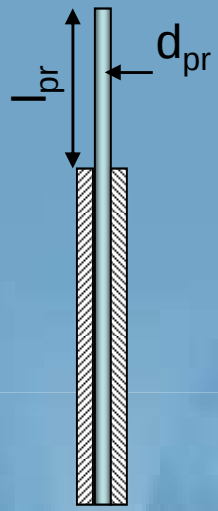
As a matter of fact, probes (“sounding electrodes”) were used in the late 19th and early 20th centuries only to measure the voltage distribution in gas discharges.

# Typical shapes of Langmuir probes

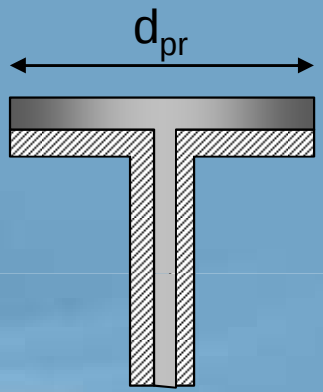
Spherical




Cylindrical




Planar (flat)

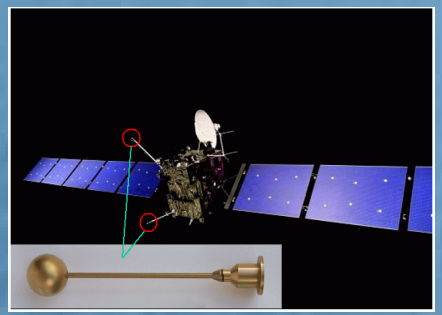



Materials

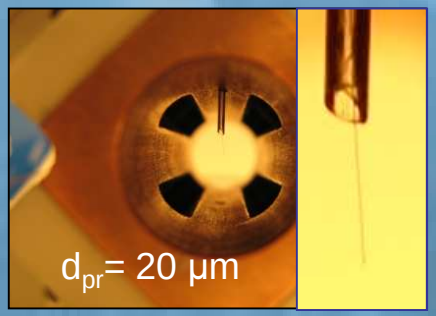
Probe: tungsten (W)
nickel (Ni)
titanium (Ti)
platinum (Pt)
molybdenum



Isolation: glass
ceramics



Ti probe (d=5cm) mounted on the body of the spacecraft. Mission “Rosetta”*

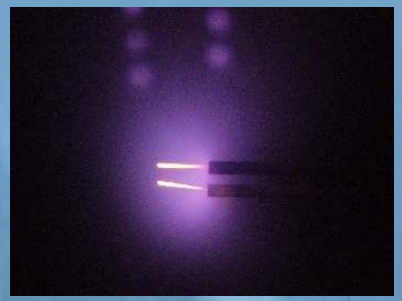



The tungsten cylindrical probe



Planar probe for ion flux measurements, with guard ring

www.electricpropulsionlaboratory.com



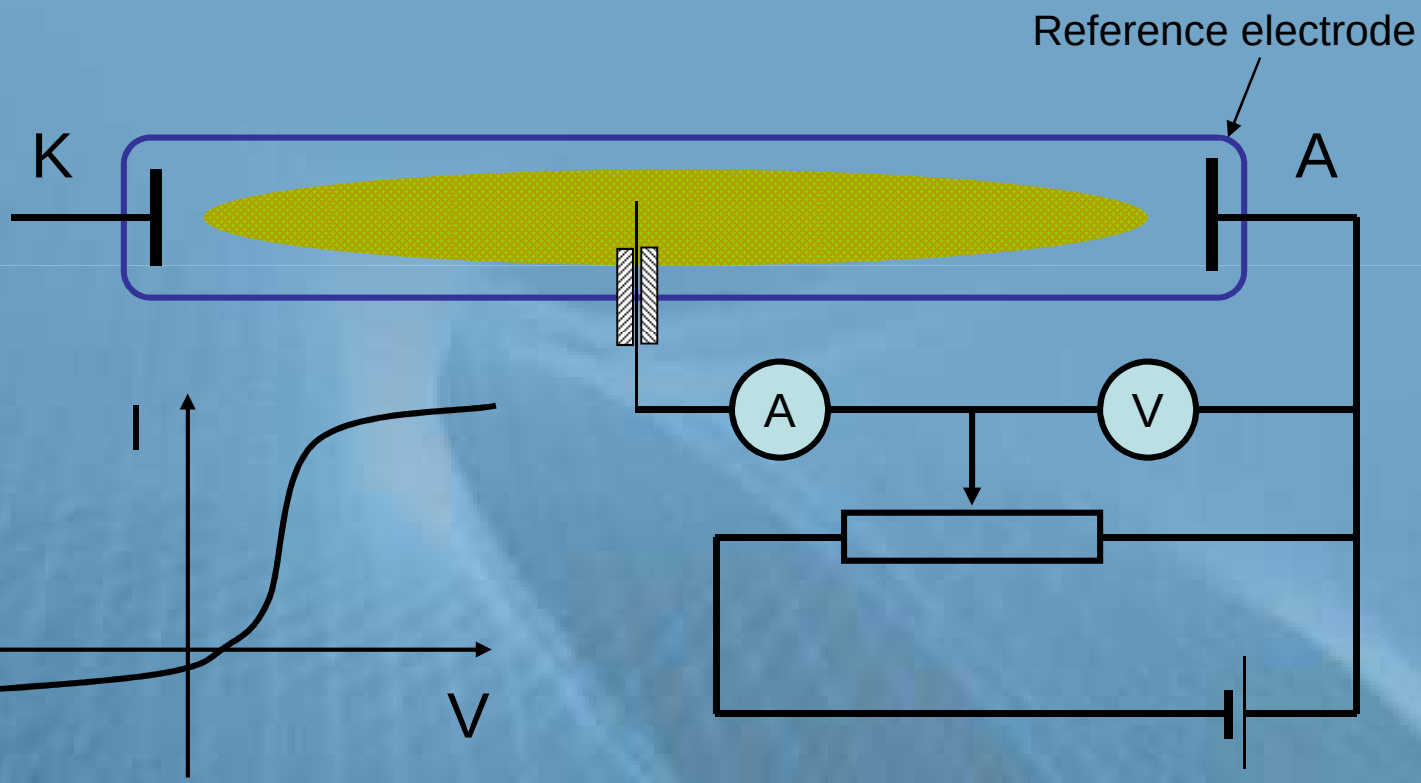
Double Langmuir probe

www.plasmaconsult.com

*Trotignon, J. G., *et al.*, *Adv. Space Res.*, 24, No. 9, 1149-1158, June 1999.

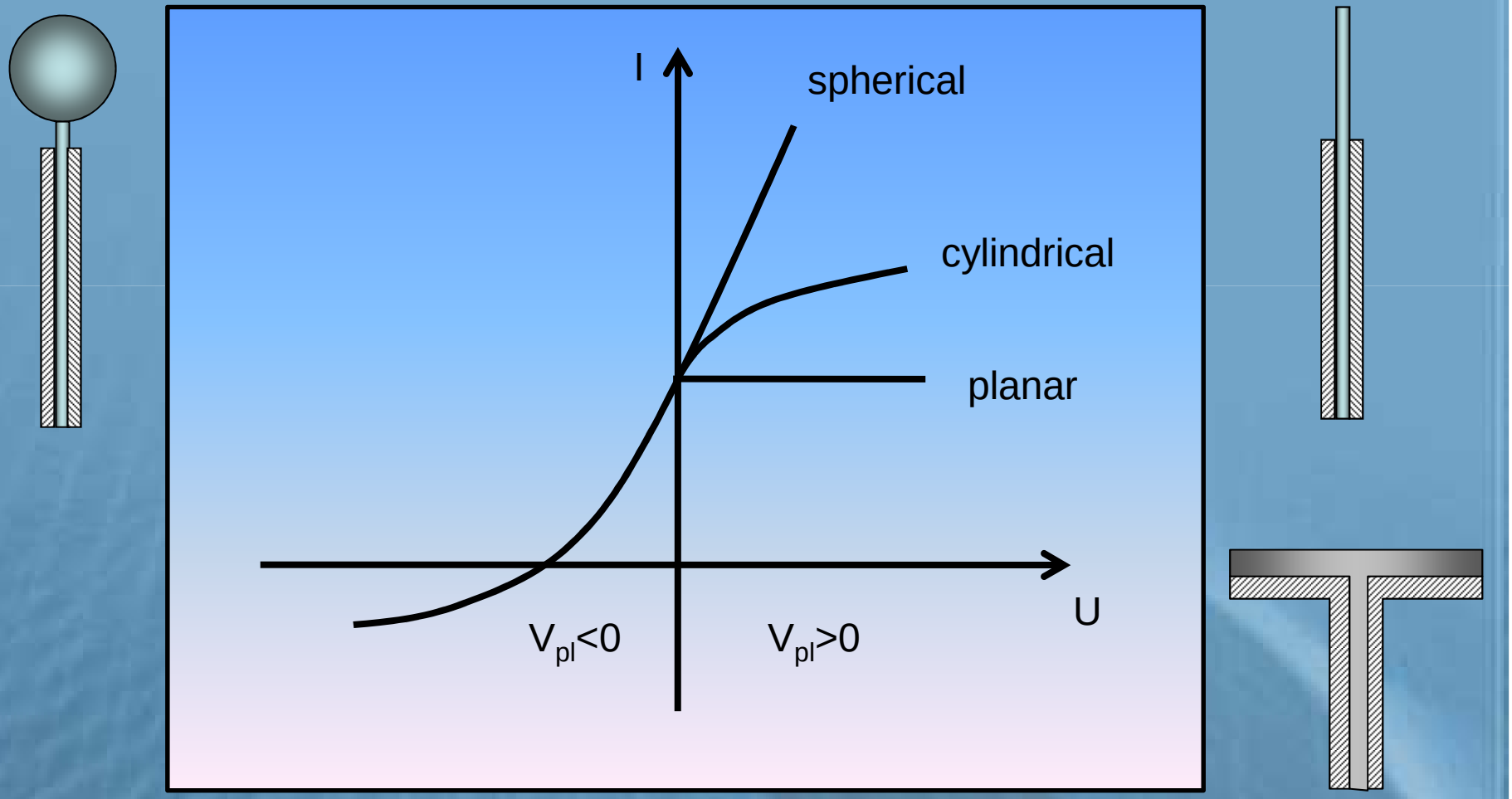
# Basics of probe diagnostic

Typical probe circuit

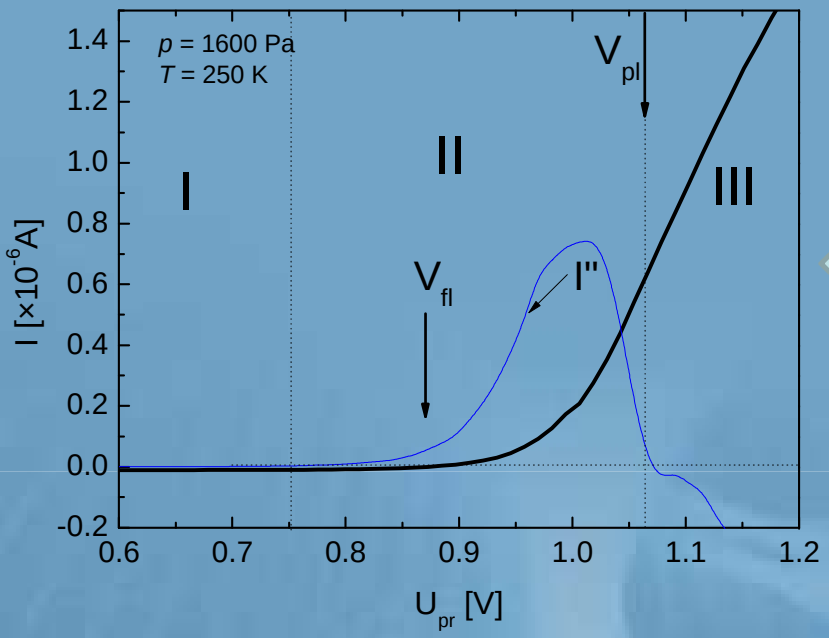



Current-voltage characteristic

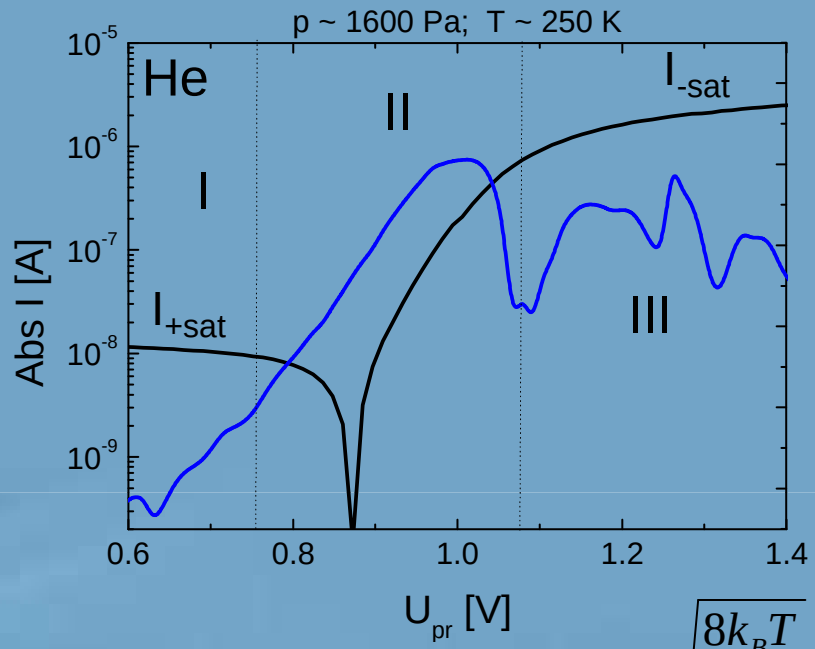
# Ideal I-V characteristics



# The probe characteristic




log




$V_{pl}$ – plasma potential (space potential)

$V_{fl}$ – floating potential

$I_{+sat} < I_{-sat} \rightarrow m_e < m_i$

$$\upsilon_{th} = \sqrt{\frac{8k_B T}{m\pi}}$$

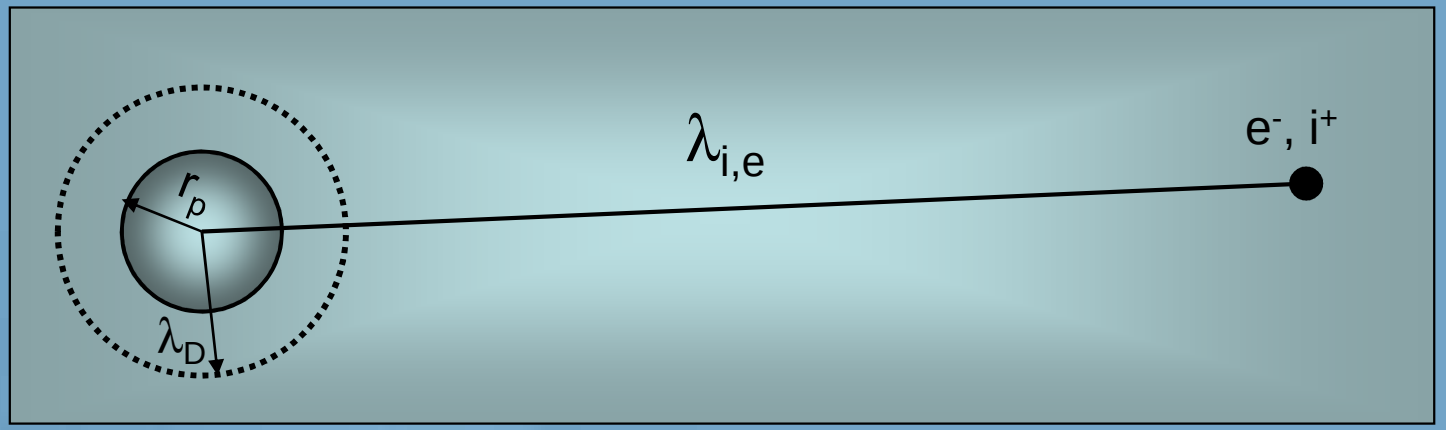
For Ar plasma $I_{-sat}/I_{-sat} \approx 250\text{-}300$

| | |
|---|---|
| I – positive ion acceleration region | $\lvert I_+ \rvert >> \lvert I_- \rvert$ |
| II – transition region | $\lvert I_+ \rvert \approx \lvert I_- \rvert$ |
| III – electron acceleration region | $\lvert I_+ \rvert << \lvert I_- \rvert$ |

$V_{fl}$ – $I_+$ and $I_-$ compensate each other
$I_{pr}$ -equals to zero

$V_{pl}$ can be determined from I" method

# Working regimes of Langmuir probe




$\lambda_D$ Debye shielding length $\Rightarrow$ $\lambda_D = \sqrt{\frac{\varepsilon_0 k_B T_e}{q_0^2 n_e}}$ $\Rightarrow$ $\lambda_D = \sqrt{\frac{\varepsilon_0 k_B}{q_0^2 n_e}\frac{T_i T_e}{T_i + T_e}}$

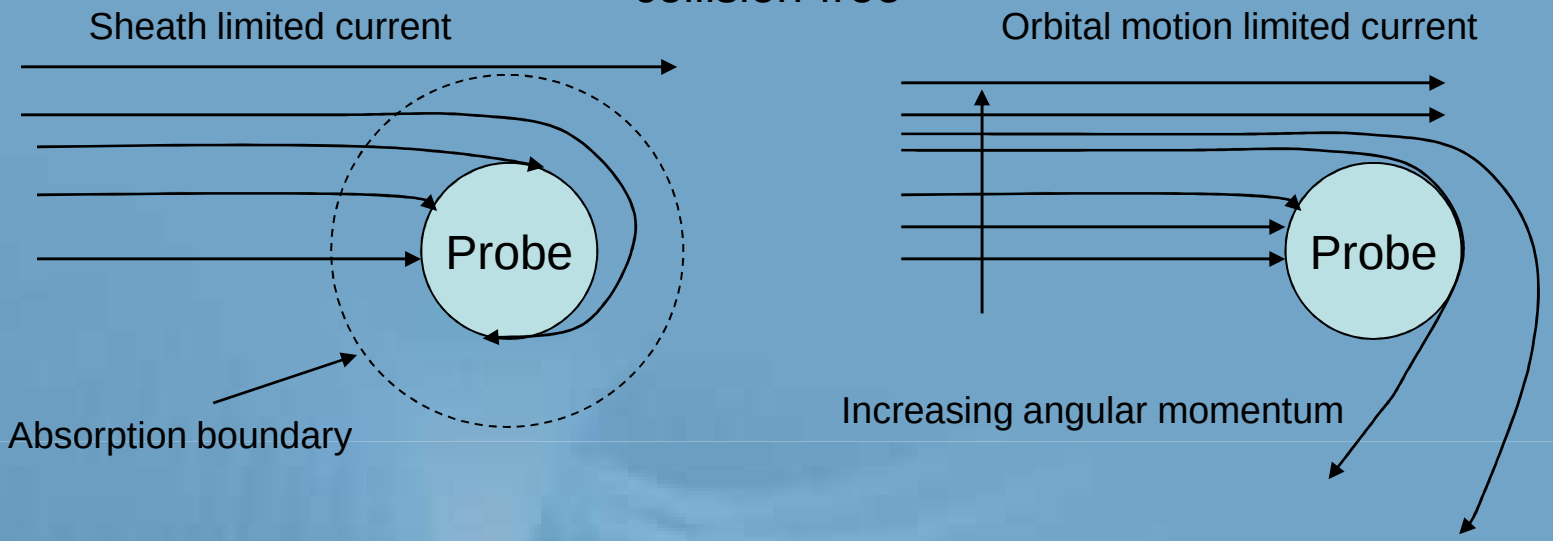
$r_p$ probe dimension

$\lambda_i$, $\lambda_e$ Mean free path for ions, electrons

| | |
|---|---|
| $\lambda_{i,e} >> \lambda_D >> r_p$ | Collision-free movement of ions and electrons in thick sheath (orbital motion limit, OML -> Langmuir theory). |
| $\lambda_{i,e} >> r_p >> \lambda_D$ | Collision-free movement of charge particles in thin sheath (space charge limit). |
| $\lambda_D >> \lambda_{i,e} >> r_p$ | Probe current is determined by collisions of charged and neutral particles in space-charge sheath around the probe. ~~EEDF~~ only $T_e$ |

# Orbital motion limit (OML)

collision-free




The plasma outside the sheath is assumed to be perfectly neutral and the sheath boundary is well-defined

Ion current ~~dependence~~ Electron current

$r_p = 10^{-2}$ cm; $\lambda_e \sim 10^{-1}/p$ [cm]; $\lambda_i \sim 10^{-2}/p$ [cm].

For $n_e = 10^{10}$-$10^{12}$ cm$^{-3}$; $T_e$=1eV; $\lambda_D \sim 10^{-2}$-$10^{-3}$ cm$^{-3}$

$\Rightarrow \lambda_{i,e} >> \lambda_D >> r_p \Rightarrow$ $P_i$ <<1 Torr, $P_e$<< 10 Torr $\Rightarrow$ <1Torr

# Orbital motion limit (OML)

collision-free

$r_p = 10^{-2}$ cm; $\lambda_e \sim 10^{-1}/p$ [cm]; $\lambda_i \sim 10^{-2}/p$ [cm].

For $n_e = 10^{10}$-$10^{12}$ cm$^{-3}$; $T_e$=1eV; $\lambda_D \sim 10^{-2}$-$10^{-3}$ cm$^{-3}$

⇨ $\lambda_{i,e} >> \lambda_D >> r_p$ ⇨ $P_i$ <<1 Torr, $P_e$<< 10 Torr ⇨ <1Torr

⇩

$\lambda_i << \lambda_e$

At very low pressures p → $\lambda i,e >> \lambda_D$

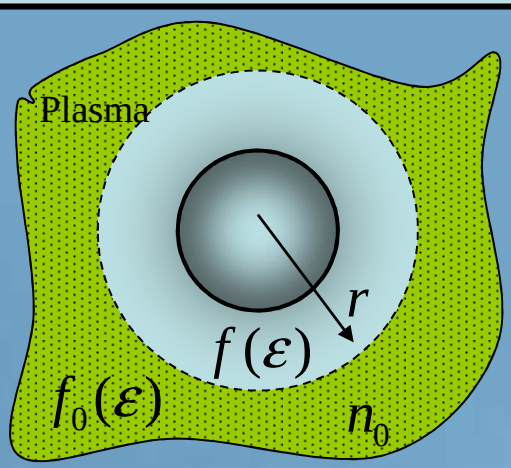
„Transition“ region of pressures → $\lambda_e > \lambda_D \geq \lambda_i$.

- Ions: collision case
- Electrons: collisionless case

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In high-density plasmas it is impossible to set the probe voltage up to the plasma potential in order to determine the electron density. However, the plasma density can be also estimated from the ion current part of the probe characteristic. Therefore a great effort has been devoted to the theoretical description of the positive ion collection by an electrical probe.

# Orbital motion limit (OML)




The electric field caused by the difference between the potential of the probe and the plasma potential at the place where the probe is located is limited to the volume inside the probe sheath boundary and does not penetrate behind it.

**Charged particle in the retarding field**

$f(\varepsilon)$ - EEDF electron energy distribution function

$n_0$ - Plasma density $S$ - Probe area

$$f(\varepsilon) = f_0(\varepsilon + eV(r))$$

current density ⟹

$$J_e(V) = \frac{2\pi e n_0}{m^2} \int_{eV}^{\infty} f_0(\varepsilon)(\varepsilon - eV)\, d\varepsilon$$

In case of a spatially homogeneous plasma ⟹ $I_e = J_e(V)S$

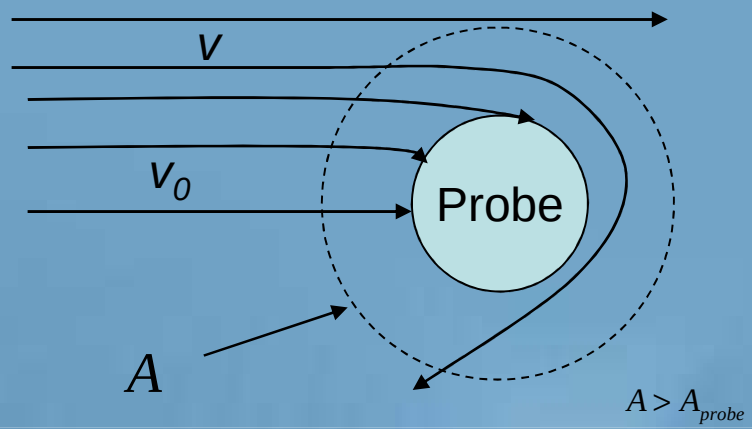
$$V = V_{probe} - V_{pl}$$

Maxwellian EEDF $f_0(\varepsilon) = \left(\frac{m}{2\pi k T_e}\right)^{3/2} e^{\left(-\frac{\varepsilon}{kT_e}\right)}$ ⟹

$$I_e(V) = I_{e0} \exp\left(-\frac{eV}{kT_e}\right)$$

$$I_{e0} = \frac{1}{4} e n_0 \sqrt{\frac{8kT_e}{\pi m}}$$

# Orbital motion limit (OML)




## Attracting probe

For the interval of charged particle velocities $[0,v_0]$ the probe acts as a collector with the effective collecting area $A$ that is greater than the true surface area of the probe. For charged particle velocities a part of charged particles $v>v_0$ which entered the probe sheath passes by the probe and is never collected by the probe (so-called Orbital Motion Limited current, OML).

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Bohm criterion* (sheath formation) $T_i<T_e$

**probe potential is screened only to** $k_B T_e / 2q_0$

Consequence: the positive ions are accelerated in the pre-sheath and the ion current is determined by Te.

**The case A → ∞ has been solved by Langmuir****

$$I_+(V) \approx 2qn_0 S\sqrt{\frac{2qV}{m_i}}$$

**Cylindrical probe**

**The case A → 0, Bohm equation**

$$I_+(V) \approx 0.4qn_0 S\sqrt{\frac{2qT_e}{m_i}}$$

$T_i$ is not present

*D. Bohm, et al. The Characteristics of Electrical Discharges in Magnetic Fields, A. Guthrie, R.K. Wakerling, Eds., McGraw-Hill, New York 1949 p.77.

**I. Langmuir, H.M. Mott-Smith, Phys. Rev. 28(1926)727.

# Theories of ion collection. Collisionless case

## Finite sheath and $E_i > 0$

**Spherical probe, monoenergetic ions**

I.B. Bernstein, I. Rabinowitz, Phys. Fluids 2(1959)112

**Spherical and cylindrical probe, $E_i$ monoenergetic and Ei Maxwellian**

**B**ernstein, **R**abinowitz, **L**aframboise (BRL) theory

J.G. Laframboise, UTIAS Report 100(1966)

I.B. Bernstein, and I.N. Rabinowitz, Phys. Fluids **2**, 112 (1959).

**A**llen-**B**oyd-**R**eynolds (ABR) theory

J.E. Allen, R.L.F. Boyd, and P. Reynolds, Proc. Phys. Soc. (London) B70, 297 (1957). (spherical probe)

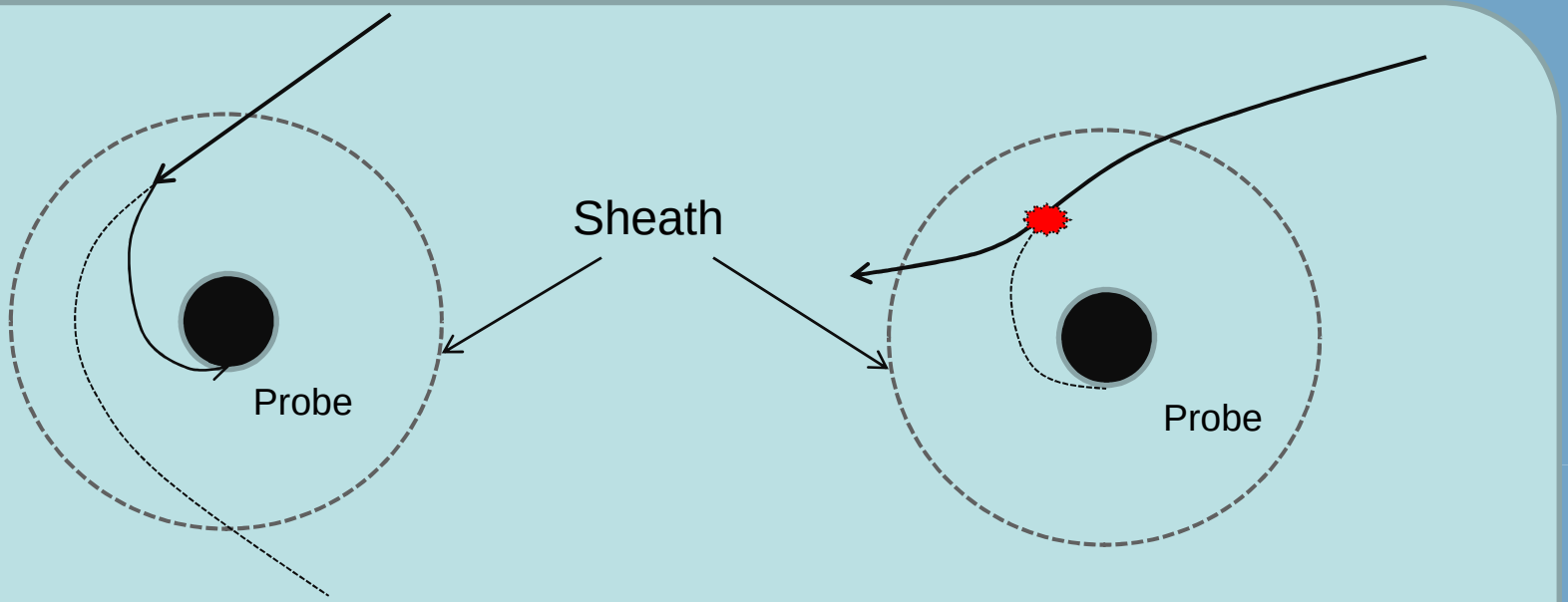
F.F. Chen, J. Nucl. Energy, Pt. C 7, 47 (1965). (cylidrical probe)

**General model for $E_i>0$ (electrons Maxwellian)**

J.I. Fernández Palop, J. Ballesteros, V. Colomer, M.A. Hernández, J. Phys. D: Appl. Phys. 29(1996)2832

The model includes the ABR-Chen model for $E_i=0$ as a special case.

# Influence of ion-neutral collisions




Increase ion current due to destruction of ion orbital motion

Elastic scattering cause decrease of the ion current

**Theory**

*Talbot L., Chou Y. S., Rarefied Gas Dynamics (Academic Press, 1966) p. 1723.*
*Chou Y. S., Talbot L., Willis D. R., Phys. Fluids **9**(1966) 2150.*
R.H. Kirchhoff, E.W. Peterson, L. Talbot, *AIAA Journ.* **9**(1971)1686.

**The effect of orbital motion destruction**:

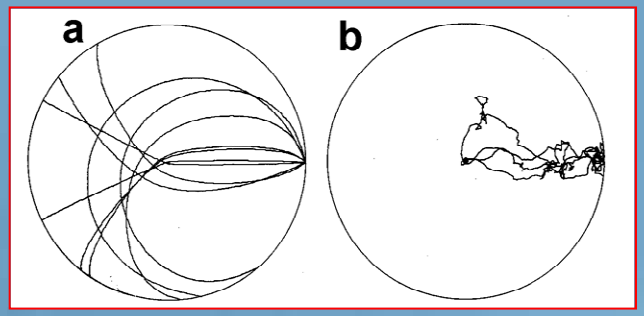
*Zakrzewski Z., Kopiczynski T., Plasma Phys. **16**(1974)1195*.

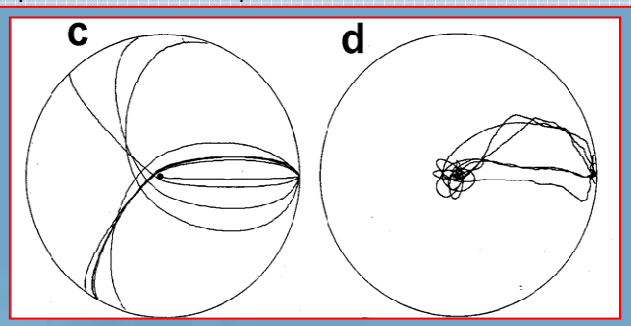
**Comparison of different theories**

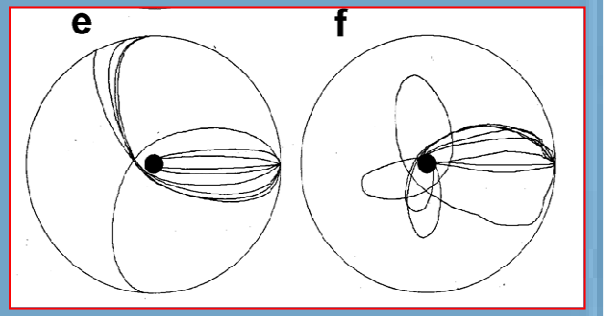
M. Tichý, M. Šícha, P. David, T. David, *Contrib. Plasma Phys.* **34**(1994)59
P. Kudrna, PhD Thesis, MFF, Charles University, Prague, Czech Republic (1996) (in Czech).

# Influence of ion-neutral collisions in the probe space charge sheath

Flowing afterglow, pressure 160 Pa, $Ar^+$ ions in He carrier gas*
$d_{pr}$ = 50 μm, $L_{pr}$ = 4 mm, $T_i$=300K



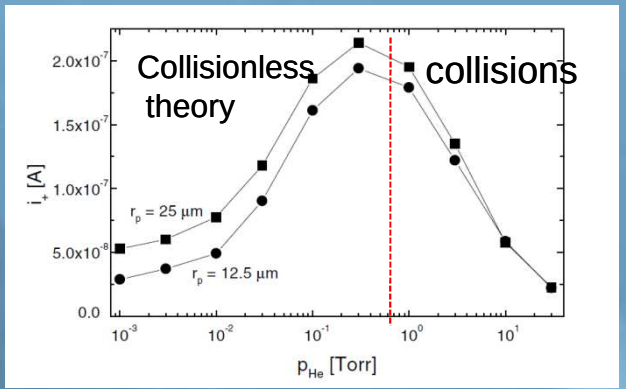

$n_e \approx 10^7$ cm$^{-3}$,
sheath thickness 6.7 mm

$n_e \approx 5\times10^8$ cm$^{-3}$,
sheath thickness 1.1 mm

$n_e \approx 10^{10}$ cm$^{-3}$,
sheath thickness 0.31 mm

**Influence of Ion – Neutral Collisions on the Ion Currents****

At lower electron number densities and higher probe voltages the sheath radius is large, it is much larger than probe radius and it depends on probe radius only very weakly. At higher pressures the ions collide in the sheath and they are captured in the potential well around the probe. Then the ion current depends on sheath radius only and does not depend on the probe radius.




The ion current versus neutral gas pressure for two probe diameters at *Up = −2 V* and $n_e = 5\times10^8$ *cm*$^{-3}$.

**Trunec, D., Španěl, P., Smith, D., Contrib. Plasma Phys. 35(1995)203*

**D. Trunec, P. Spanel, D. Smith Contrib. Plasma Phys. 42 (2002) 1, 91−*98*

# Determination of the electron temperature

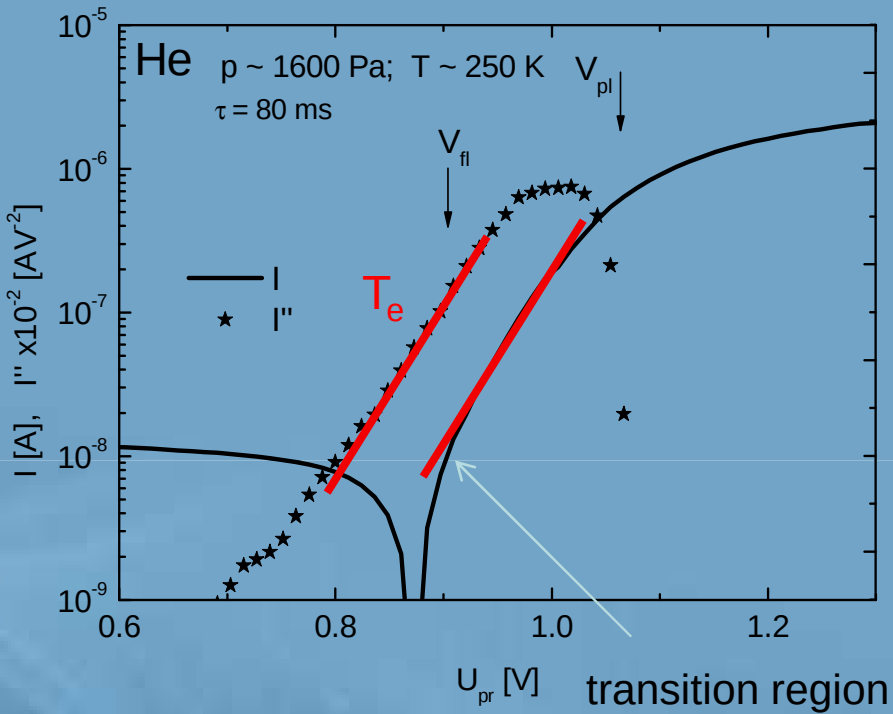
The electron temperature $T_e$ for any plasma is well defined if the EEDF is Maxwellian

$$I_e(V) = I_{e0}\exp\left(-\frac{eV}{kT_e}\right) \qquad I_{e0} = \frac{1}{4}en_0\sqrt{\frac{8kT_e}{\pi m}}$$

$$\frac{kT_e}{q} = \left[\frac{\Delta \ln(I_-)}{\Delta U_{pr}}\right]^{-1}$$

Since the derivative of an exponential function is exponential function, the slope in semilogarithmic scale of the second derivative also gives $T_e$

$$\frac{kT_e}{q} = \left[\frac{\Delta \ln(I_-^{//})}{\Delta U_{pr}}\right]^{-1}$$




We need $I_-$ only!!!

$I_-$???

Real: $I(U_{pr})=I_-+I_+$

# Determination of the electron temperature

It is usually assumed that the ion current can be written as*:

$$I_i = I_{i0}(1+\eta)^{\chi}$$

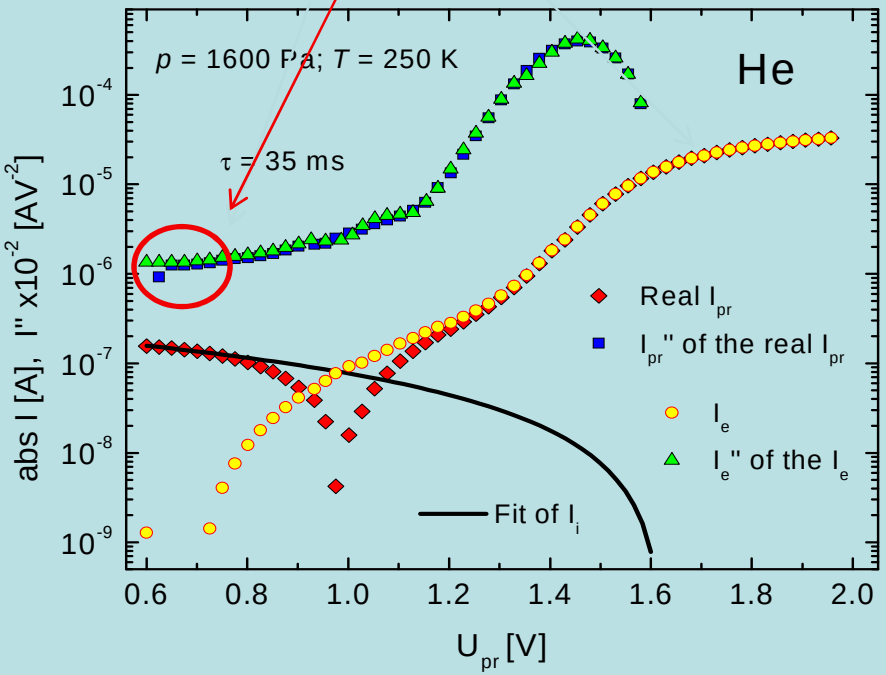
$I_{io}$ and $\chi$ are the unknown fitting parameters, which can be estimated by means of an experimentally determined dependence of the saturated ion current $I_i$. $\eta$ is the normalized dimensionless probe potential in the electron retarding region:

$$\eta = \frac{q(U_{pr} - V_{pl})}{k_B T_e}$$

$$I_e = I_{pr} - I_i$$

*B. Nuhn, G. Peter, Proc. XIII. ICPIG Berlin (1977), p. 999.

For low temperature equilibrium plasma the influence of the ion current has the same order of magnitude as noise and sometimes can be neglected




Comparison between the second derivative $I_{pr}$" (blue squares) of the measured characteristic $I_{pr}$ (red diamonds) in Helium afterglow and the second derivative $I_e$" (green triangles) of the electron current of the probe $I_e$ (yellow circles). Full curve indicates the obtained ion current $I_i$

# Determination of the electron temperature

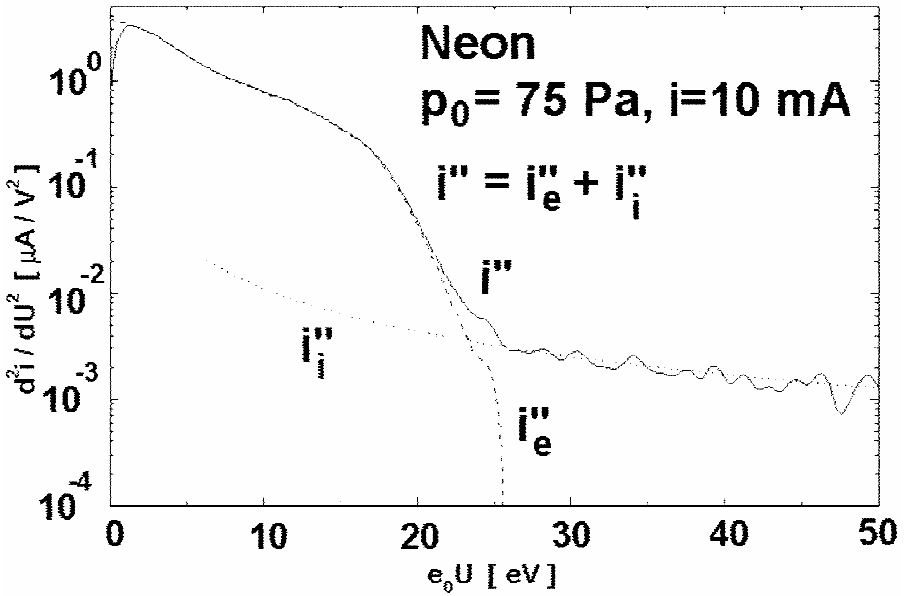
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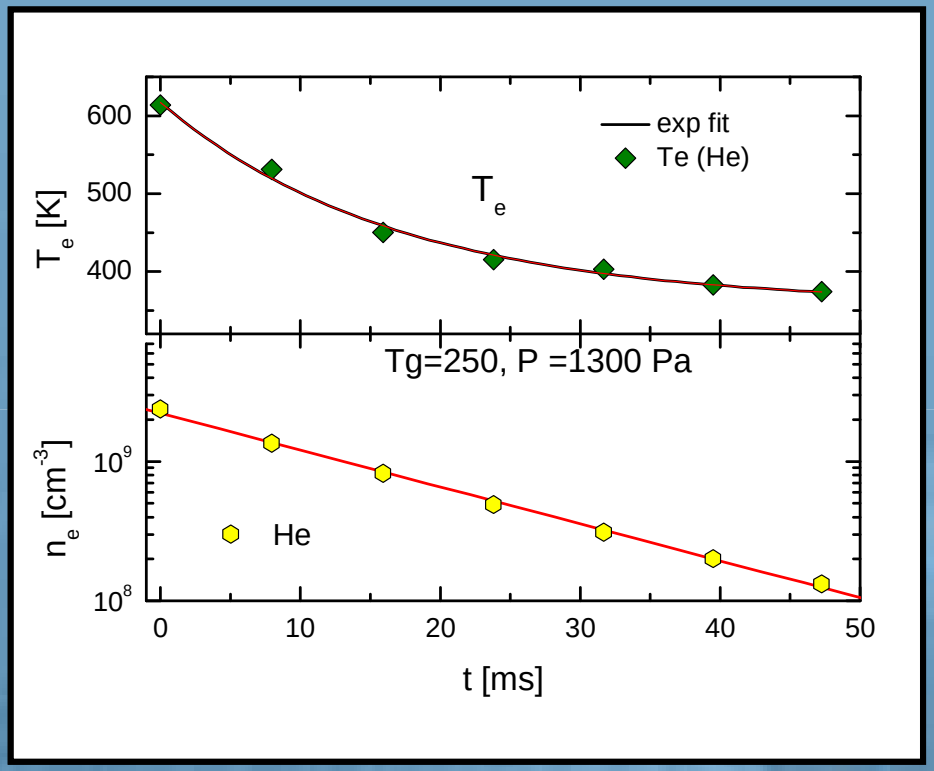
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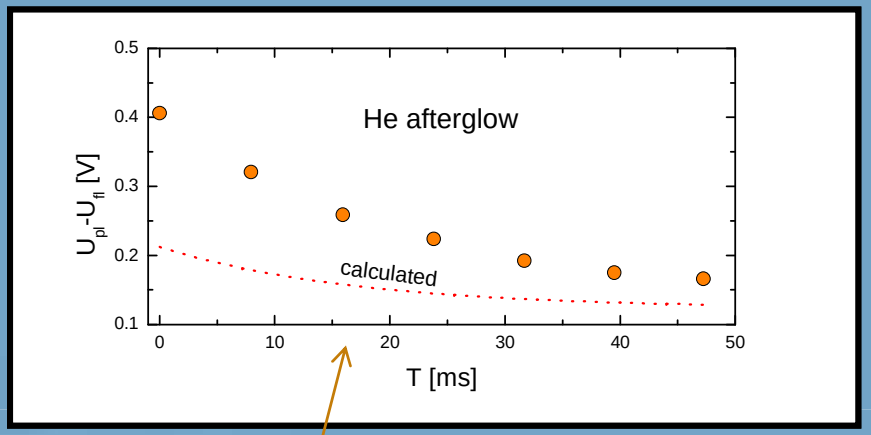



*B. Nuhn, G. Peter, Proc. XIII. ICPIG Berlin (1977), p. 999.

# Determination of the electron temperature




Upper panel. The electron temperature, Te, measured in late afterglow in He. The electron temperature is obtained from the slope of linear part of the plot of second derivative of corresponding probe characteristic. Lower panel. Measured and calculated electron density decays.




The difference of plasma and floating potentials $\Delta V = (V_{Pl} - V_f)$ in $He_2^+$ dominated plasma in He afterglow.

$$n\sqrt{\frac{T_e}{2\pi m_e}}\exp\left[\frac{-(V_p - V_f)}{T_e}\right] = 0.6\sqrt{\frac{T_e}{m_i}} \quad *$$

$$V_{pl} - V_{fl} \approx (3.3 - 0.5\ln\mu)\,T_e$$

$$\Delta V_{He}(T_e = 600\,K) \approx 0.21[V]$$

**ChinWook Cung Jr. Phys. of plasmas , **12**, 123505 (2005)*

# Determination of EEDF in plasma

Probe characteristics in comparison with other methods allows us to determine the whole EEDF.

Druyvesteyn method*

$$\int_0^\infty \bar{f}(u_p)\, u_p^{\frac{1}{2}} du_p = 1$$

$$f(E) = \frac{2}{Aq^2}\sqrt{\frac{2m_e(\mathrm{U_{pr}} - \mathrm{V_{pl}})}{q}}\,\frac{\mathrm{d}^2 I_e}{\mathrm{d}(\mathrm{U_{pr}} - \mathrm{V_{pl}})^2}$$

$E = -q(\mathrm{U_{pr}}\text{-}\mathrm{V_{pl}})$ is energy of electrons

$$f(V) \sim \sqrt{V - V_p}\,\frac{dI_e}{d(V - V_p)^2}$$

$$\frac{d^2 I}{d(V - V_p)^2} = \frac{1}{4} Aq^2 \sqrt{\frac{2q}{m_e(V - V_p)}}\,[f(E)]$$

$$\frac{d^2 I}{d(V - V_p)^2} = \frac{1}{4} Aq^3 \sqrt{2\pi m_e}\,(k_B T_e)^{-3/2} e^{\left[\frac{q(V - V_p)}{k_B T_e}\right]}$$

Maxwellian EEDF

*Druyvesteyn M.J., *Z. Physik* **64**(1930)781

**The second derivative**

**directly measured (on-line methods)**

Second harmonic generation
***G.R. Branner et al Rev. Sci. Instr. 34(1963)231***

Direct analog differentiators using operational amplifiers and sawtooth-like probe voltage
*V.A. Godyak et al., Plasma Sources Sci. Technol. 1(1992)36*

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**calculated (off-line methods)**

Based on algorithms of the numerical analysis

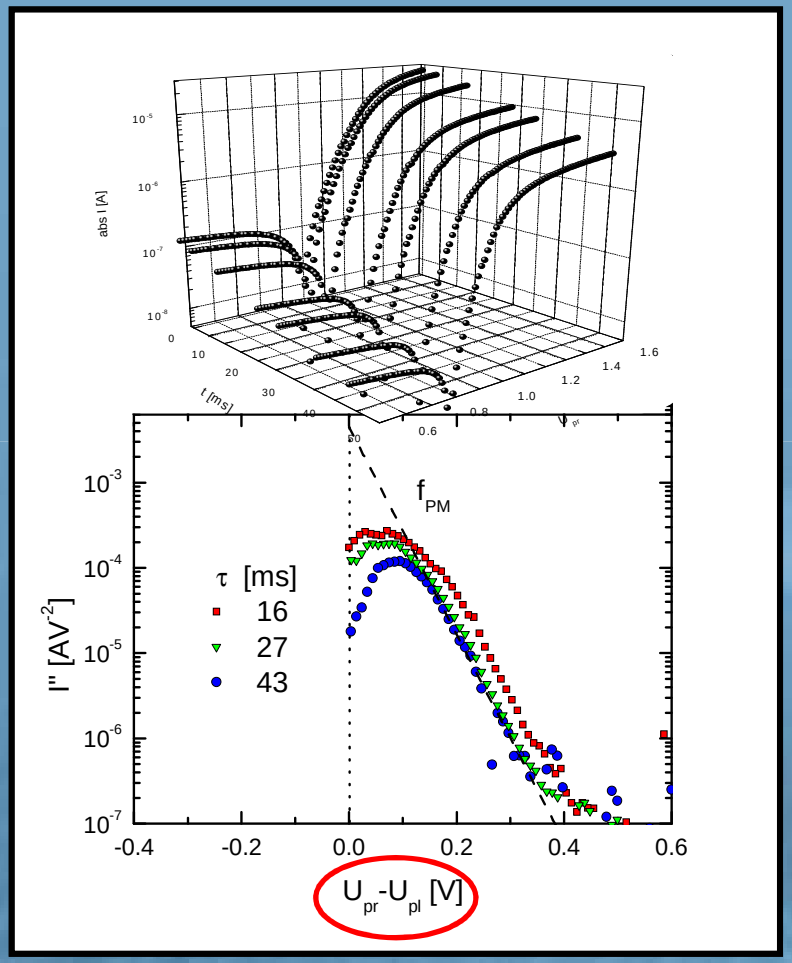
Non-recursive digital filterring
D. Trunec, Contrib. Plasma Phys. 32(1992)523,

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For noisy data most common method is S-G filtering
A. Savitzky, M.J.E. Golay, *Anal. Chem.* **36**(1964)1627

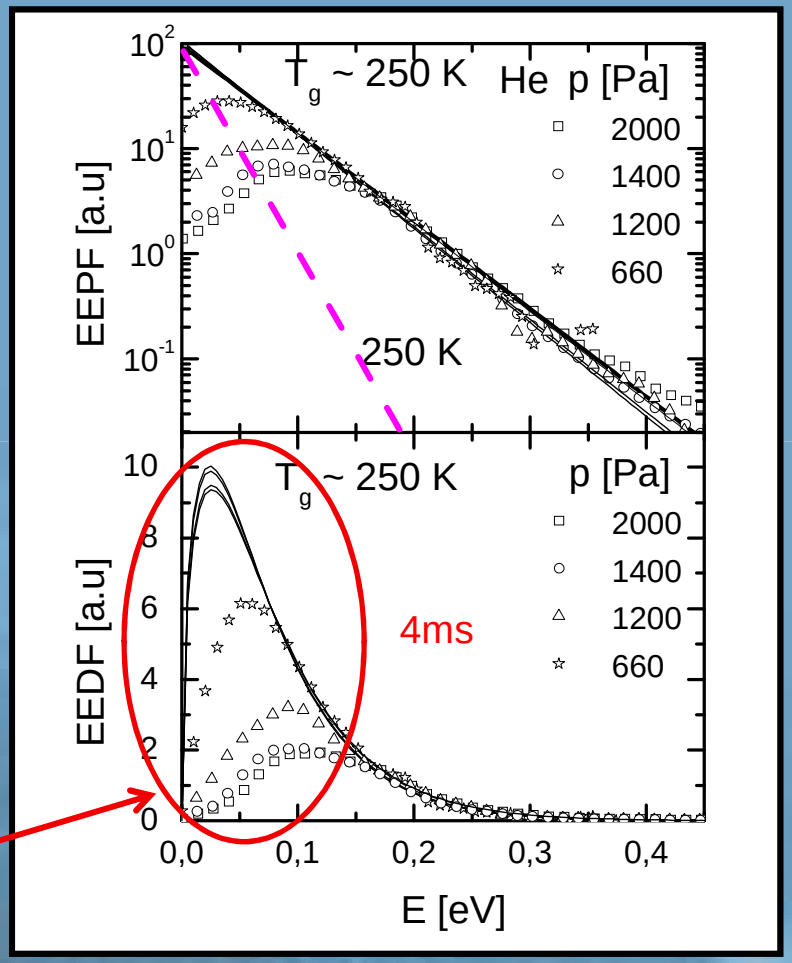
$V_p$ ⟹ from the zero crossing of I"

# Determination of EEDF in plasma




The measured probe characteristics and corresponding second derivatives.

???




The EEDFs in helium afterglow for different pressures.

$$f_{EEPF} \sim f_{EEDF}/\sqrt{E}$$

I. Korolov, R. Plasil, T. Kotrik, P. Dohnal, O. Novotny, and J. Glosik, Contrib. Plasma Phys. **48, 461 2008.**

# Correction

The Druyvesteyn method is successful when the free path of electrons is mach lager than the probe radius. The high pressure significantly shortens the free path of the electron. For these gas pressures, Swift* where pointed out the distortion of the result from I" probe measurements due to plasma depletion caused by electron sink on the probe surface.

*Swift J D and SchwarMJR 1970 Electrical Probes for Plasma Diagnostics (London: Iliffe Books)

## Correction I

$$f(eV) = \frac{3m^2\gamma_0}{8\pi e^3}\left( j_e''(eV) - \frac{\psi_0}{V} j_e'(eV) \right)$$

$$\psi_0 = \frac{ac_i}{\gamma_0\lambda}$$ diffusion parameter

*Gorbunov, N. A., et al., Plasma Sources Sci. Technol.* ***11*** *(2002), 492 – 497*

## Correction II

$$I''(U_S) = Cf_p(qU_S) - C\int_{eUs}^{\infty} K''(\varepsilon, qU_S) f_P(\varepsilon)\,d\varepsilon$$

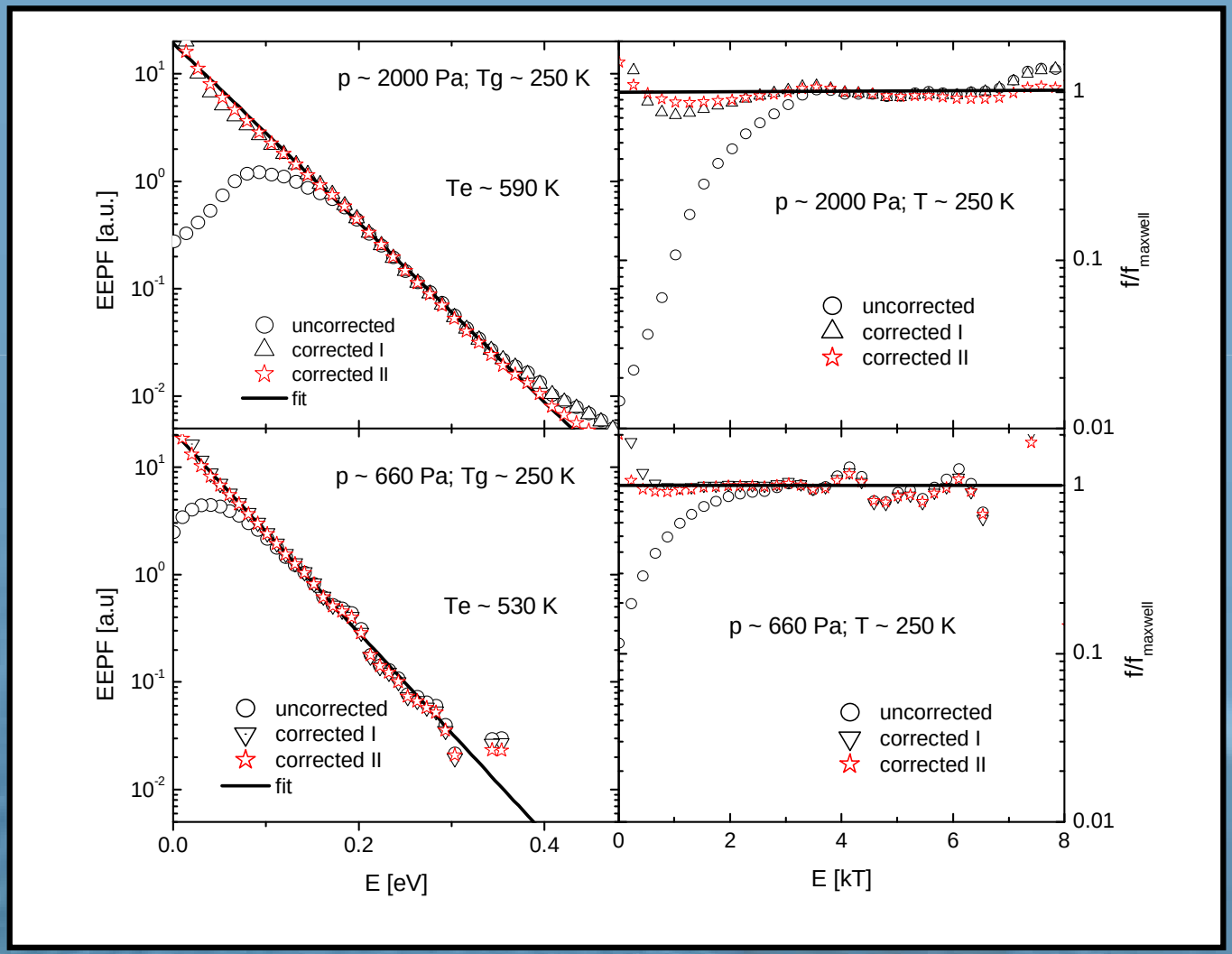
$$K''(\varepsilon,U) = \frac{2\psi\varepsilon^2}{[\varepsilon(1+\psi)-\psi eU]^3},\quad C = \frac{ne^3S}{2\gamma\sqrt{2m}}$$

$\gamma$ = geometrical factor

$$\psi = \frac{r_p \ln\left(\frac{\pi l_p}{4r_p}\right)}{\gamma\lambda}$$

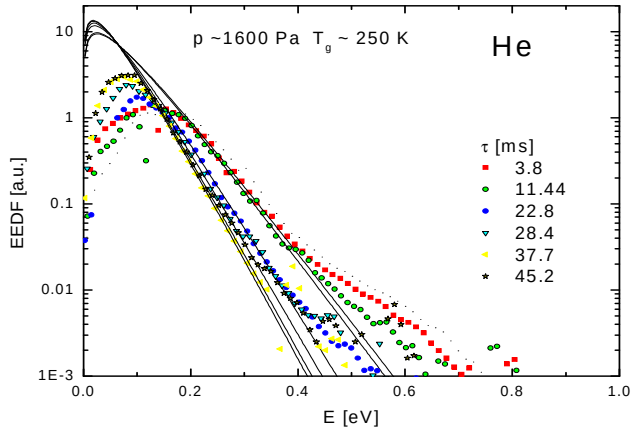
*Dimitrova, M.,et al., Journal of Physics: Conference Series* ***44*** *(2006), 169 – 174.*

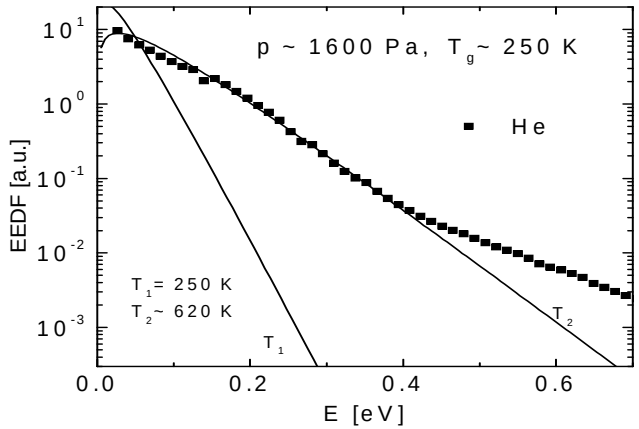
# Example of corrections



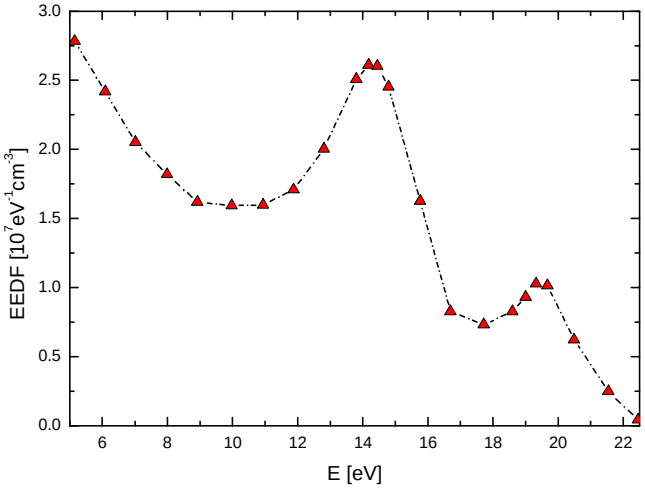

I. Korolov, R. Plasil, T. Kotrik, P. Dohnal, O. Novotny, and J. Glosik, Contrib. Plasma Phys. **48, 461 2008.**

# EEDF in helium afterglow




Measured electron energy distribution functions (EEDF) in the He plasma. Line indicates Maxwellian fit




Measured high energy portion of the EEDF in a helium 130 Pa afterglow plasma under conditions of self-trapping of fast electrons *.

$$He(2^3S) + e \longrightarrow \varepsilon(19.8eV)$$

$$He(2^3S) + He(2^3S) \longrightarrow \varepsilon(15.05eV)$$

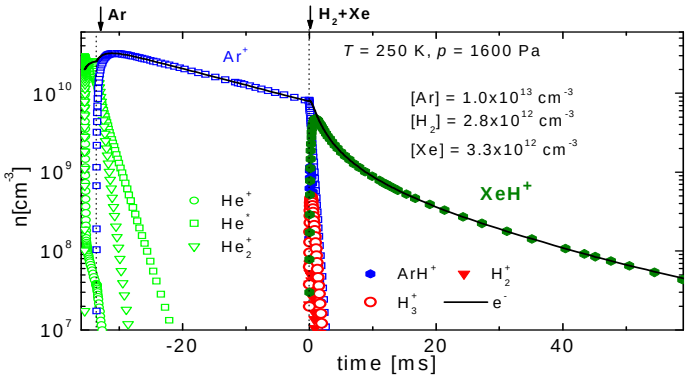
$$He(2^3S) + He(2^1S) \longrightarrow \varepsilon(15.85eV)$$

$$He(2^1S) + He(2^1S) \longrightarrow \varepsilon(16.6eV)$$

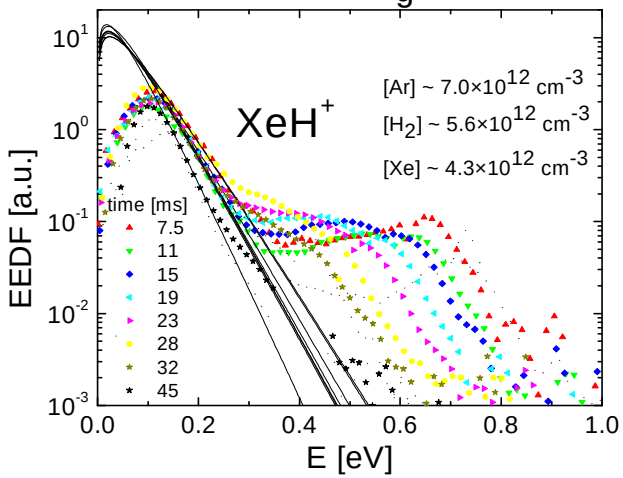
*V. I. Demidov and C. A. DeJoseph Jr. Rev. Sci. Instrum. 77, 116104 (2006)

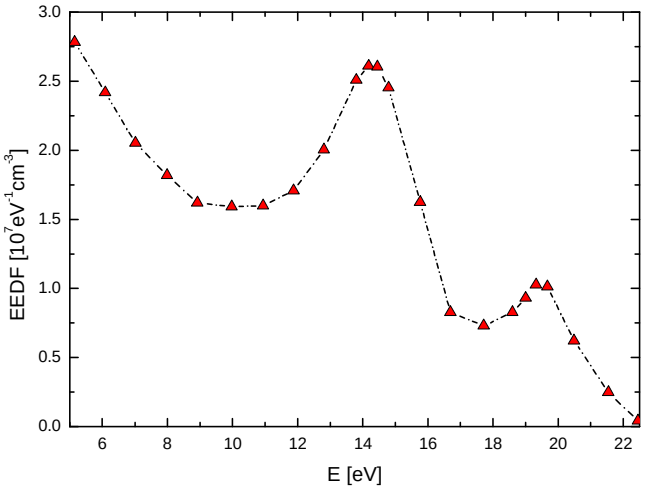
# EEDF in afterglow plasma

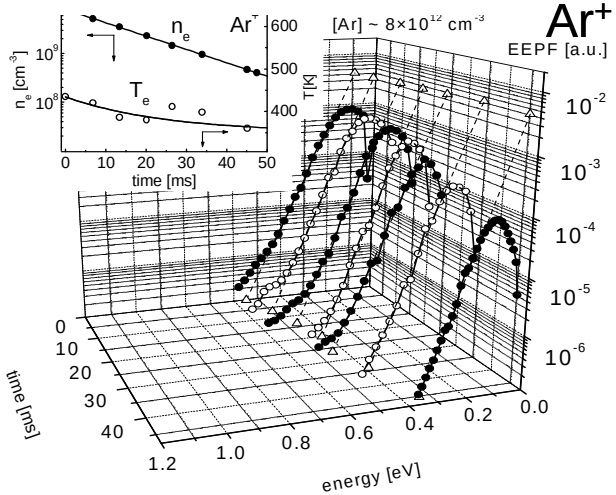
## Recombination of $XeH^+$




$p$ ~ 1600 Pa, $T_g$ ~ 250 K

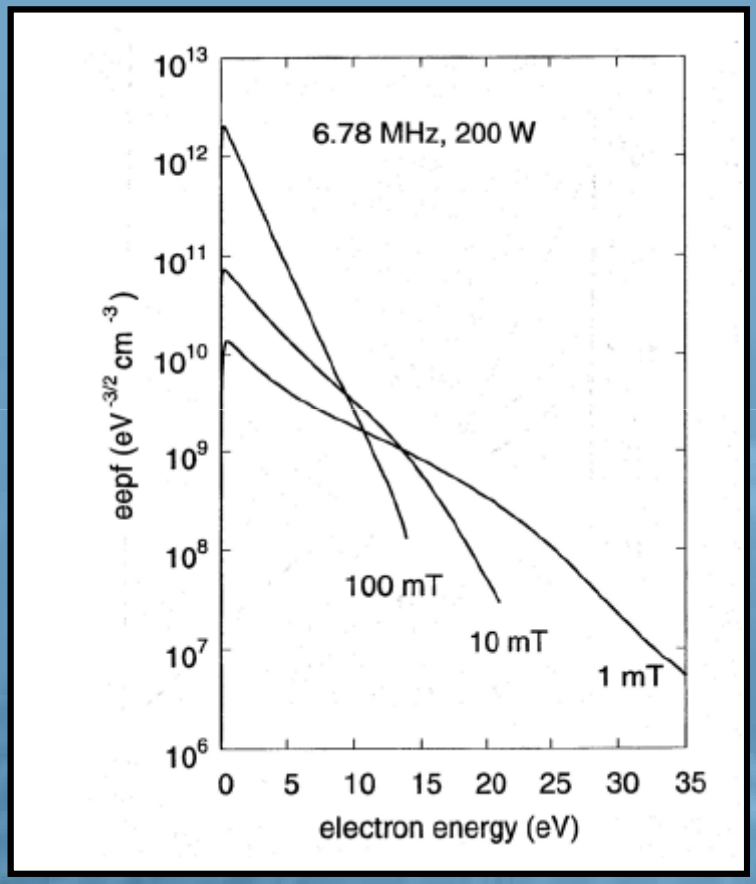


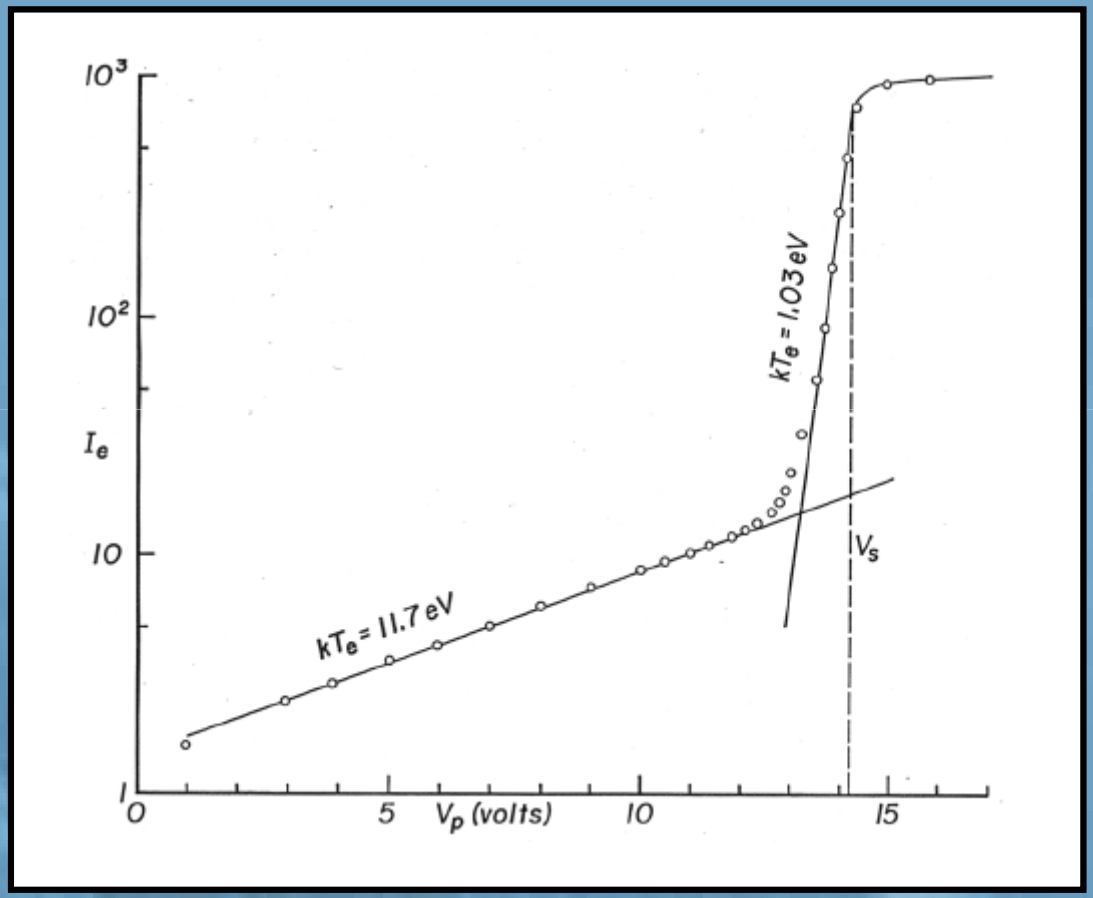




R. Plasil, I. Korolov, T. Kotrik, P. Dohnal, G. Bano, Z. Donko, and J. Glosik, Eur. Phys. J. D **54, 391 2009.**

# Bi-Maxwellian EEDF




EEDF curves obtained with a Langmuir probe in a transformer-coupled plasma (TCP) discharge




An I-V curve of a bi-Maxwellian EEDF

F.F. Chen Langmuir probe diagnostics, IEEE-ICOPS meeting, Jeju, Koreja, 2003

# Determination of electron density from probe data

## Methods

By EEDF integrating over all energies

$$n_e = \int_0^{\varepsilon_{max}} f(\varepsilon) d\varepsilon \quad \Leftrightarrow \quad n_e = \int_0^{-\infty} f(U) dU$$

The accuracy of these methods is strongly dependent on the accuracy of the plasma potential estimation.

From $I_e$

$$n_e = \frac{1}{S_{probe}\, q} \sqrt{\frac{2\pi m_e}{kT_e}}\, I_e (U_{pr} = V_{pl})$$

$V_{pl}$ ! $T_e$!

J. D. Swift and M. J. R. Schwar. *Electrical probes for plasma diagnostics. London Iliffe books Ltd., 1970.*

***I-squared*** method

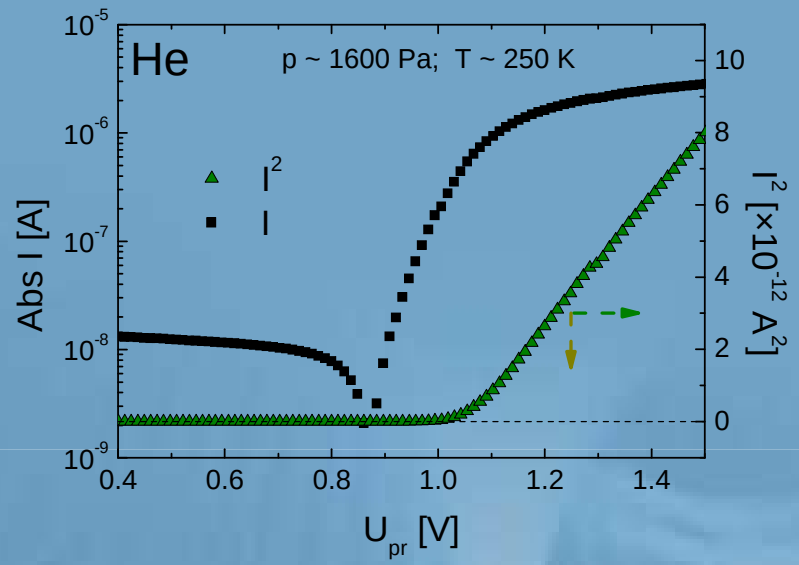
$$I^2 = \frac{2(Aqn_e)^2}{\pi^2 m_e}[k_B T_e + q(U_{pr} - V_{pl})] \qquad n_e = \sqrt{\frac{\pi^2 m_e}{2A^2 q^3} S}$$

Slope of $I^2(U)$

Španěl P.: *Int J. Mass Spectrom and Ion Proces.*, 149/150, 299, 1995

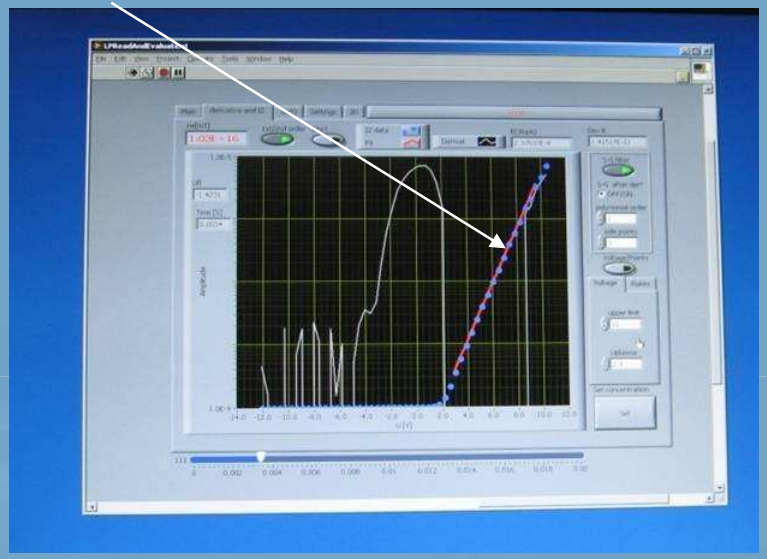
The concentration of electrons can be easily obtained without the need to determine accurate values of $V_{pl}$ and $T_e$

# I-squared method

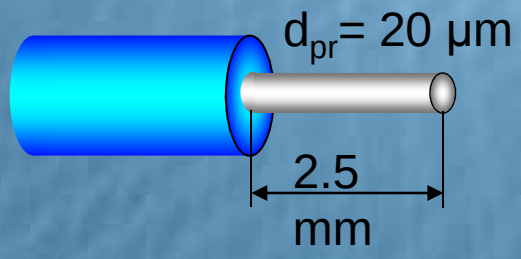



Example of measured probe characteristic (I) and its $I^2$ in Helium at 1600 Pa

slope of the linear part of $I^2$



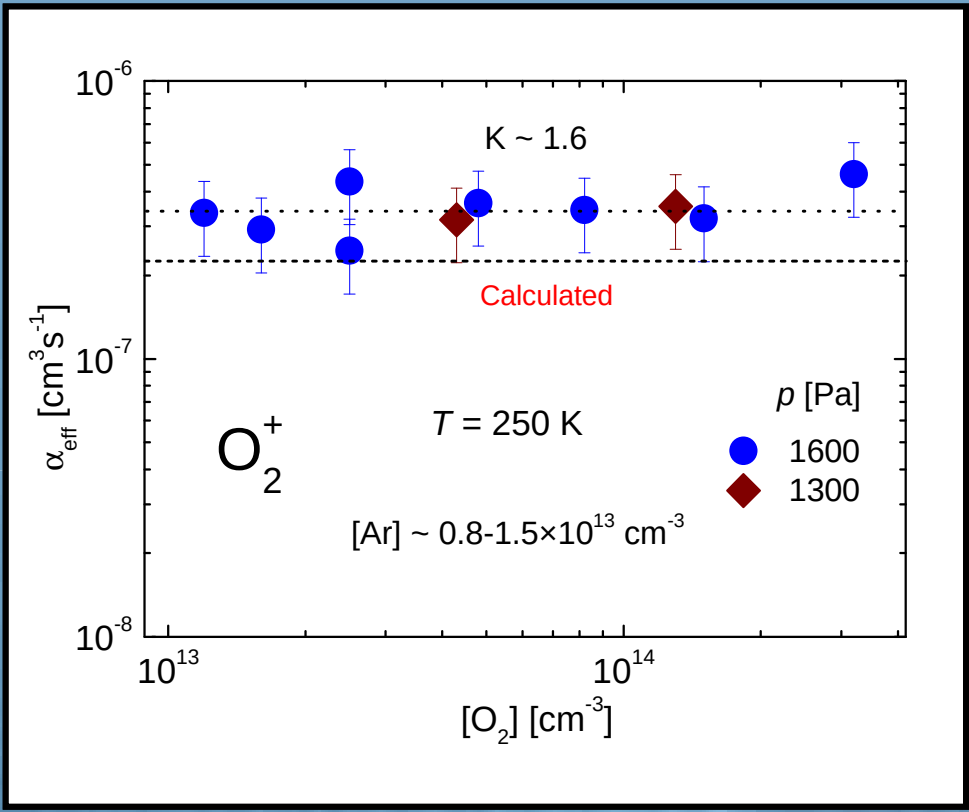
Langmuir probe configuration

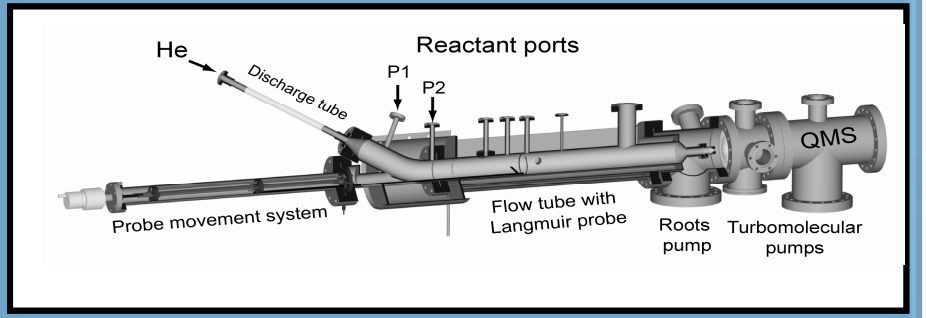



$$I^2 = \frac{2(Aqn_e)^2}{\pi^2 m_e}[k_B T_e + q(\mathrm{U_{pr}} - \mathrm{V_{pl}})]$$

$$n_e = \sqrt{\frac{\pi^2 m_e}{2A^2 q^3} S}$$

# Calibration of Langmuir probe




$$\alpha_{O_2}(T) = 2.0\times10^{-7}(300/T)^{0.65}\ \mathrm{cm^3 s^{-1}}$$

$$n_e(t) = \frac{n_0}{\alpha\tau_{loss} n_0 (e^{t/\tau_{loss}} - 1) + e^{t/\tau_{loss}}}$$

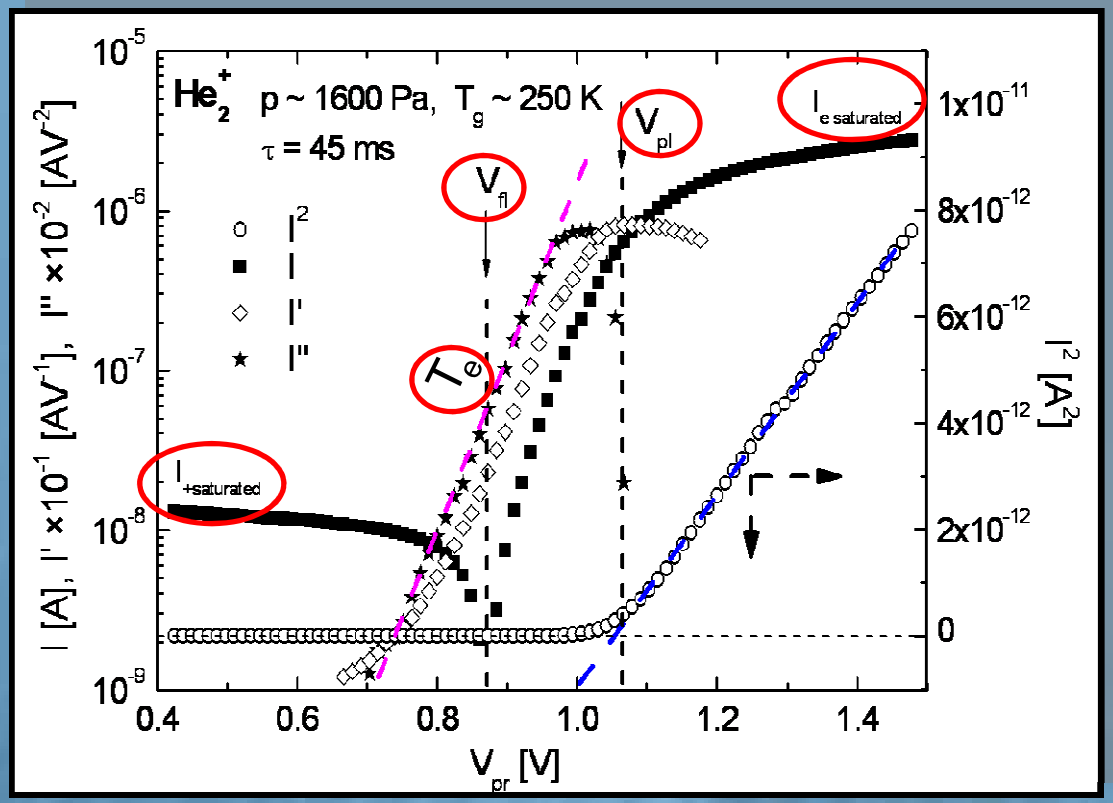
Typical correction factor K is 1.0-1.1

Calibration of the Langmuir probe. The effective recombination rate coefficients has been measured in He-Ar-$O_2$ plasma using the FALP apparatus at two different pressures of the buffer gas and for different concentrations of $O_2$. The values of the obtained recombination rate coefficient differ from the expected value $\alpha$(250 K) = $2.25\times10^{-7}$ $cm^3s^{-1}$ (red dashed line) of the recombination of $O_2^+$ ions; K = 1.6 is the applicable correction factor.


I. Korolov, O. Novotny, R. Plasil, P. Hlavenka, T. Kotrik, M. Tichy, P. Kudrna, J. Glosik and A. Luca, *Czech. J. Phys.* **56** (2006), p. B854.

V. Poterya, J. Glosik, R. Plasil, M. Tichy, P. Kudrna and A. Pysanenko, *Phys. Rev. Lett.* **88** (2002), p. 044802

# Probe characteristics. Summary




$$I_p^2 \rightarrow n_e$$

$$\frac{d^2 I}{d(V-V_p)^2} = \frac{1}{4} A q^2 \sqrt{\frac{2q}{m_e (V-V_p)}} [f(E)]$$

$$f(V) \sim \sqrt{V-V_p}\, \frac{dI_e}{d(V-V_p)^2}$$

$V_{pl}$ $V_{fl}$ $T_e$

Measured probe characteristic *I*, its first *I*' and second derivative *I*'' and square $I^2$. Standard way using absolute values of probe current and derivatives is used in the plots. Indicated is plasma potential ($V_{Pl}$), floating potential ($V_{fl}$), and section of "saturated" electron and ion current. The plasma is thermal with $k_B$Te ~ 0.03 eV, that is why the probe characteristic is so narrow.

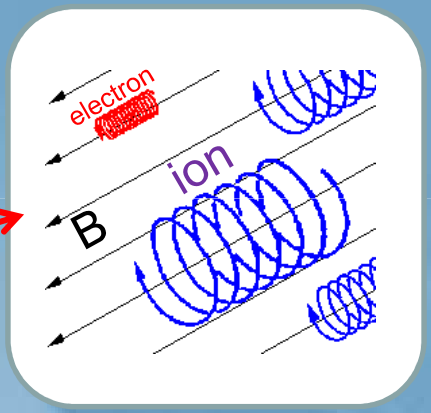
# Langmuir probe in a magnetic field

Homogeneous magnetic field

**Lorentz force** $\vec{\mathbf{F}} = e\left(\vec{\mathbf{v}} \times \vec{\mathbf{B}}\right)$

**Gyroradius** $r_g = \frac{m\,\upsilon_\perp}{|q|\,B}$

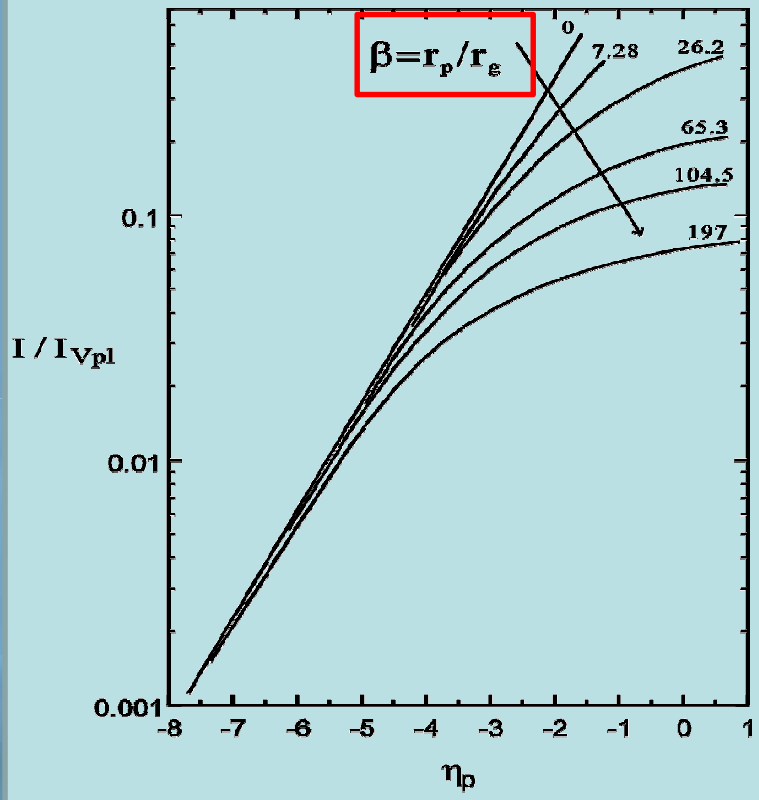
$$\bar{r}_g = \frac{1}{|q|B}\sqrt{\frac{\pi m_{e,i} k T_{e,i}}{2}}$$




$$\eta_p = \frac{e(U_{probe} - V_{plasma})}{k_B T_e}$$

**Ions are less influenced by a Magnetic Field**

**At low β only electrons are affected; ions are uninfluenced.**




The influence of the magnetic field characterised by the parameter $\beta$ on the spherical probe electron current

*Sanmartin, J.R., Phys.Fluids 13, 1 (1970)*

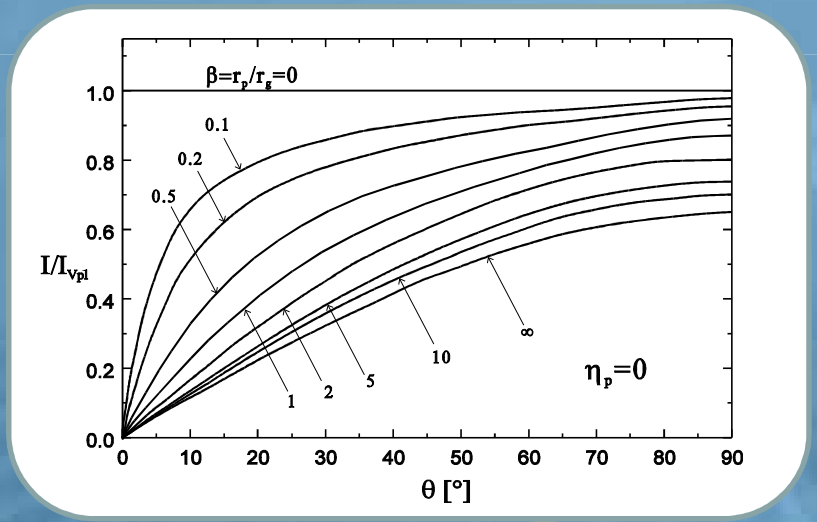
P.M. Chung, et al., Electrical Probes in Stationary and Flowing Plasmas, Theory and Application, Springer-Verlag, Berlin, Heidelberg, New York, 1975

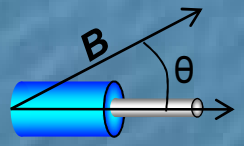
# Langmuir probe in a magnetic field

$\beta = r_p / r_g$

| | | |
|---|---|---|
| $\beta << 1$ | *weak magnetic field* | *influence can be neglected* |
| $\beta \approx 1$ | *medium magnetic field* | *small corrections* |
| $\beta > 1$ | *strong magnetic field* | *only part of $I(U_{pr})$ can be interpreted* |
| $\beta >> 1$ | *very strong magnetic field* | *uninterpretable $I(U_{pr})$* |

TOKAMAK

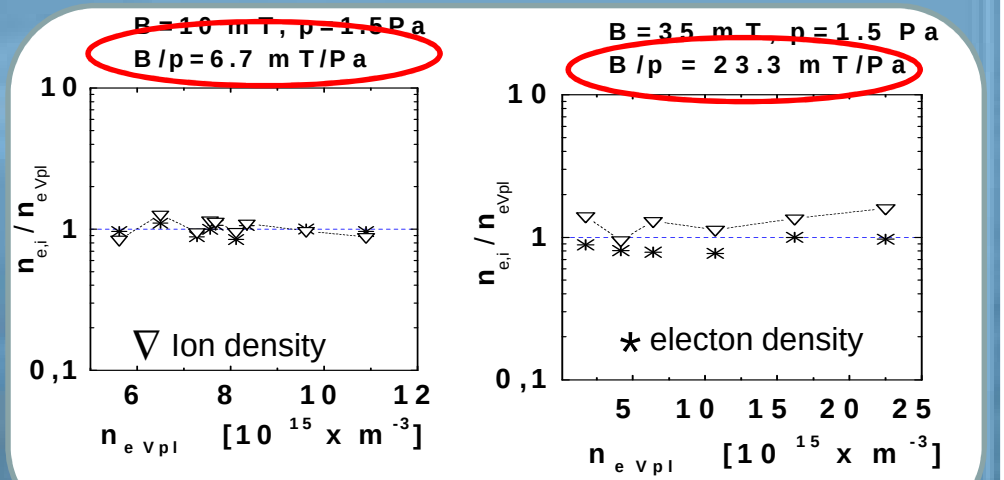



Small effect for **B**⊥Probe

Laframboise J. G. and Rubinstein Phys. Fluids 19, 1900 (1976)

DC cylindrical magnetron discharge in Ar




Kudrna, P., Passoth, E., Contrib. Plasma Phys. 37, 417 (1997)

Angular dependence of the floating potential in a magnetized plasma

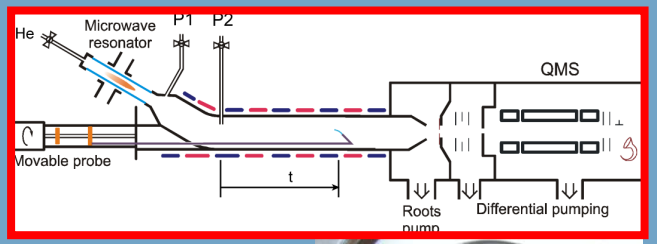
B. Koch et al., Journal of Nuclear Materials 313–316 (2003) 1114 (**flat probe**)

# Resolution of Langmuir probe

## Spatial resolution

Debye length $\lambda_D$ ⟺ Probe dimensions

spatial changes of plasma parameters




Measurements with space resolution require **movable Langmuir probe.**

200-500μm

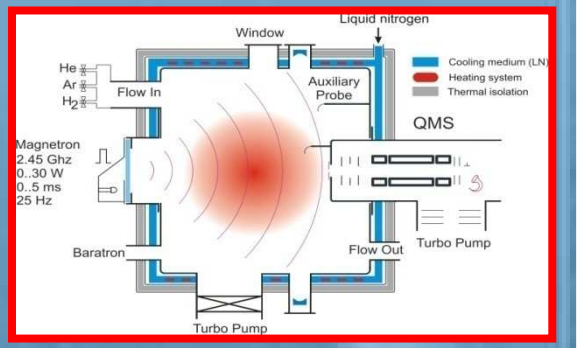


Flowing afterglow technique with Langmuir probe (FALP)

## Time resolution

Formation of a space charge sheath around the probe

Time resolution $t_r \approx 2\pi / \omega_{pi}$ $\omega_{pi}$ - ion plasma frequency

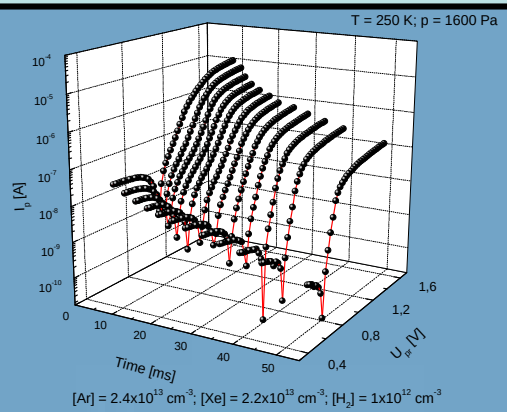


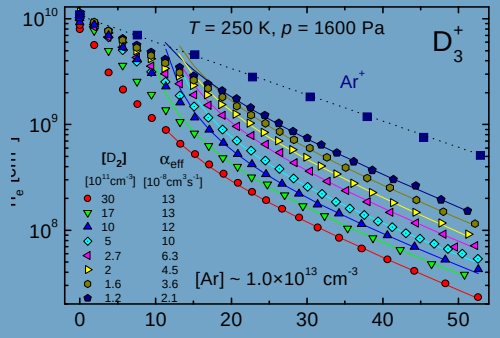

$$\omega_{pi} = \sqrt{\frac{q^2 n_i}{\varepsilon_0 m_i}}$$

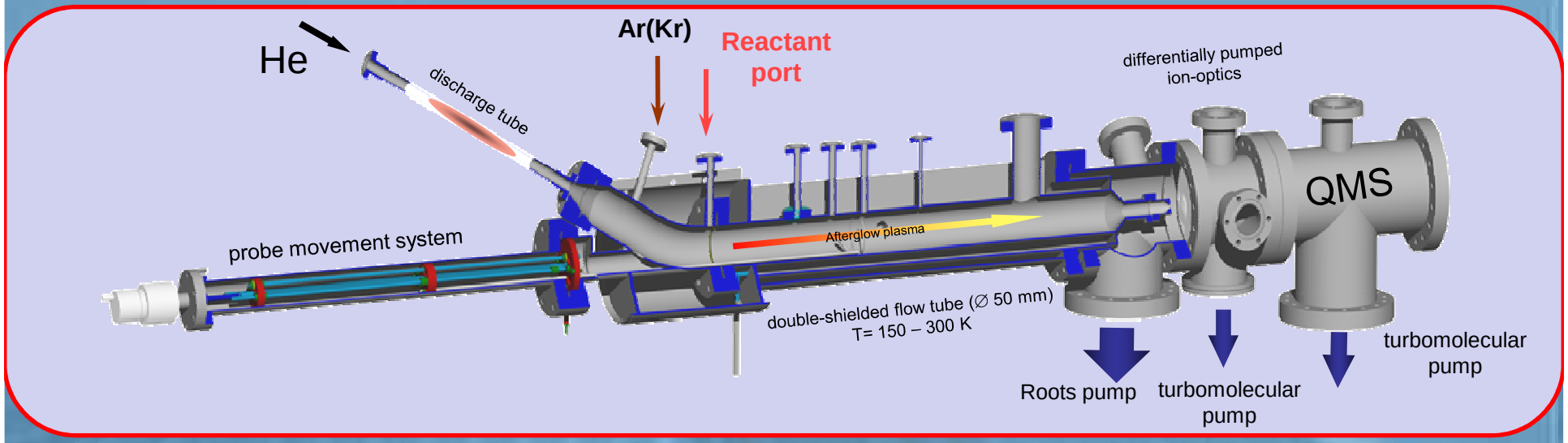
Argon: $n_i \approx n_e = 10^{10}\text{-}10^{11}$ cm$^{-3}$ $\omega_{pi} \approx 3.16\text{-}10$ MHz

$t_r \approx 0.6\text{-}2$ μs

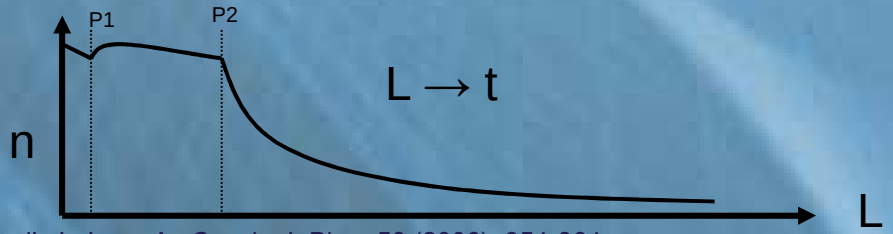
# FALP-VT. Spatial resolution of LP



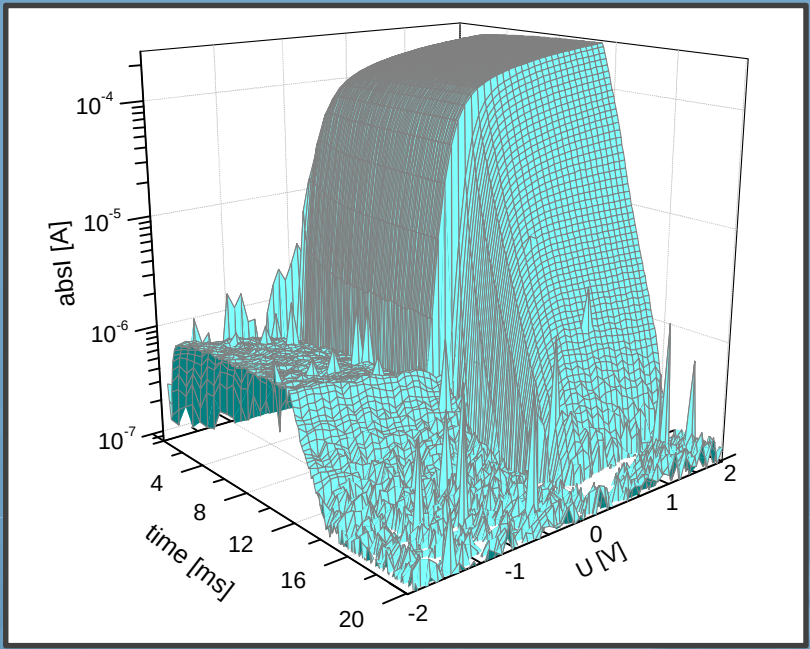




Flowing Afterglow with Langmuir Probe – Variable Temperature:
Measurable rates $\alpha > 5\times10^{-9}$ cm$^3$s$^{-1}$
Plasma decay can be monitored up to 35 cm (65)
Working temperature – 130 – 300 Pressure up to 2000 Pa

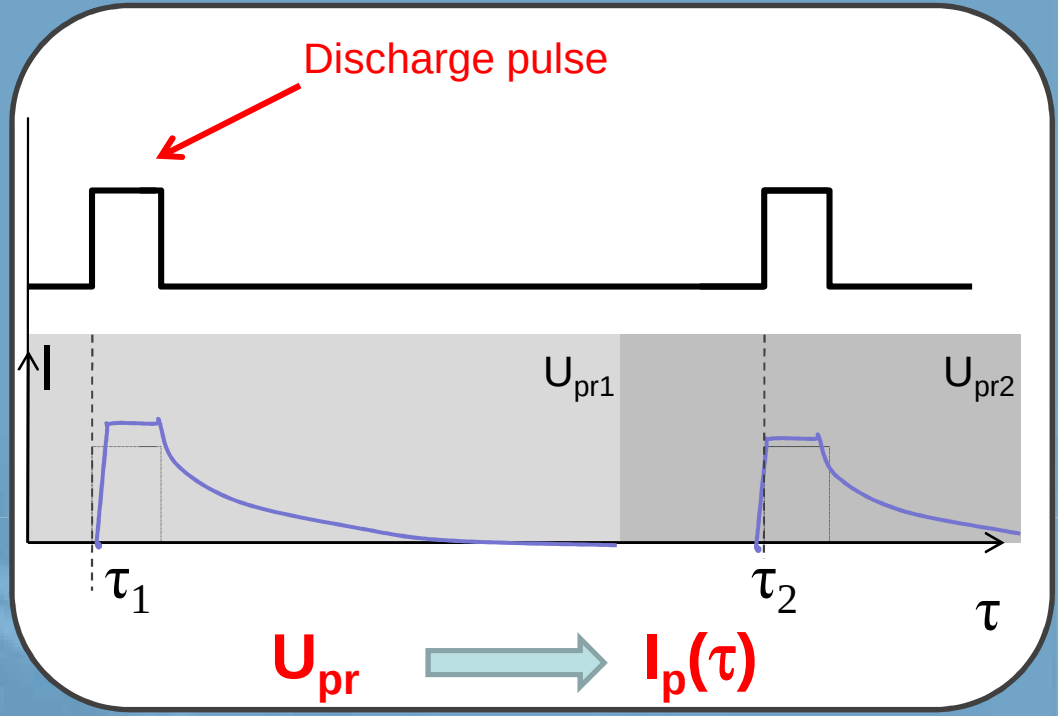



Korolov I., Novotny O., Plasil R., Hlavenka P., Kotrik T., Tichy M., Kudrna P., Glosik J., Luca A. *Czech. J. Phys.* 56 (2006) 854-864.

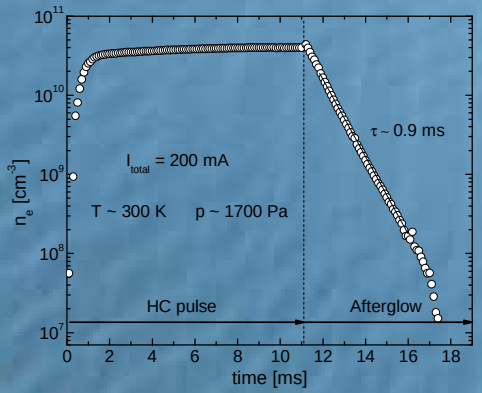
# Time-resolved method




An example of time evolution of the probe characteristic during the HC pulse and an afterglow period in He-1%Ar mixture

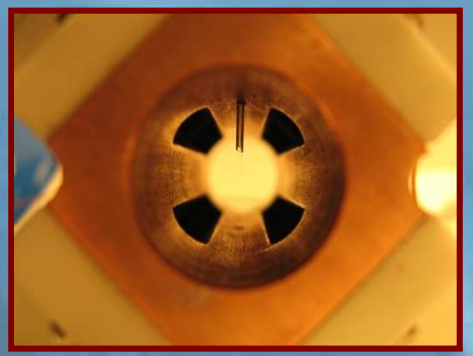



Signal sampled at a specified time - $\tau_1$, $\tau_2$ .....




Calculated electron density from the probe data as a function of time

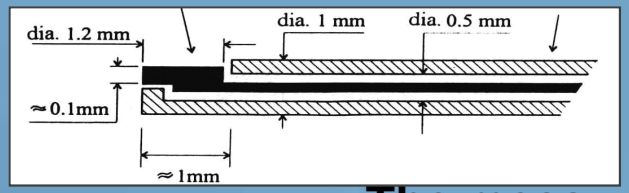
Sputtered metal source for rate coefficient measurements of asymmetric charge transfer reactions

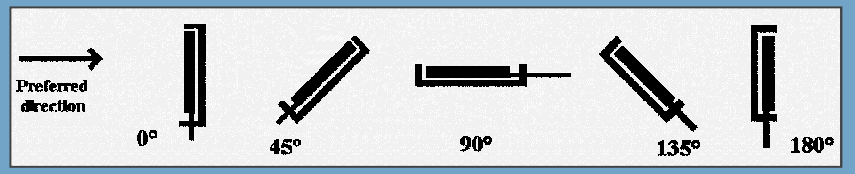


# Direction-resolved method

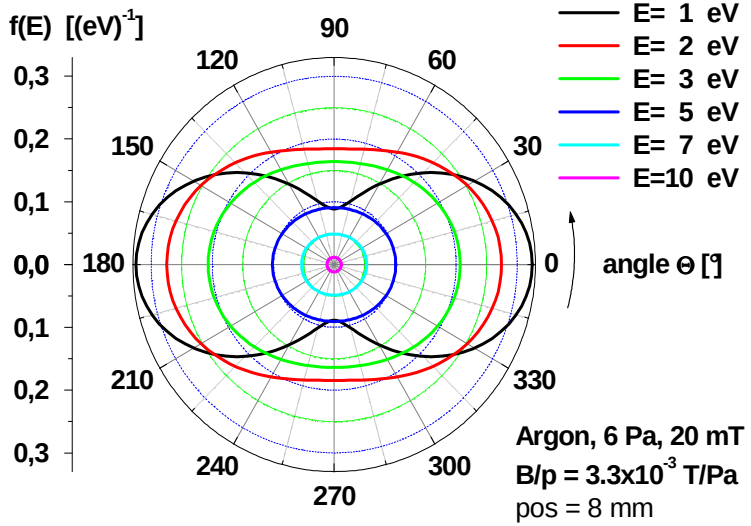
**Direction measurements: magnetic field, electric field, Etc.**

## Planar Langmuir probe

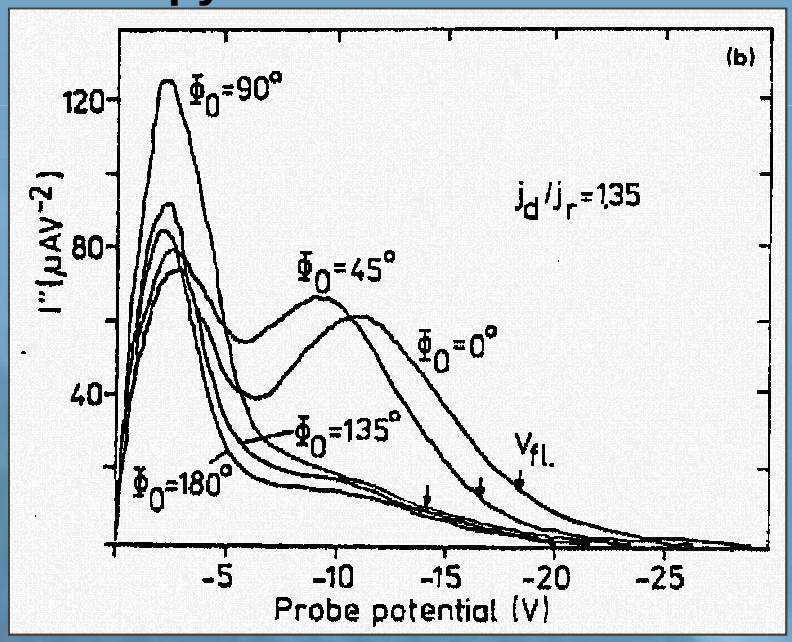



## The measurements of anisotropy of EEDF




J.F. Behnke, E. Passoth, C. Csambal, M. Tichý, P. Kudrna, D. Trunec, A. Brablec, Czechoslovak Journal of Physics B 49(1999)483-498 .



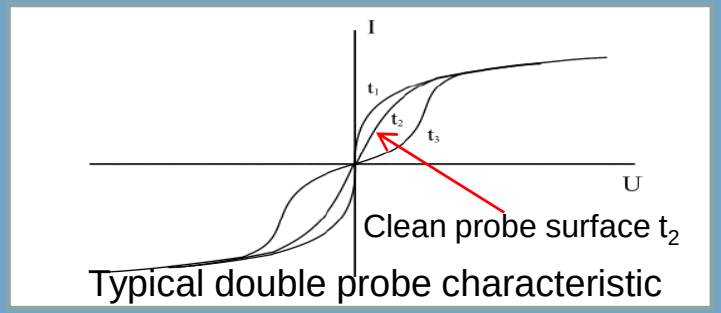

Measurements in high-current hollow cathode DC plasma jet.

S. Klagge, A. Lunk J. Appl.Phys. 70 (1991) 99

# Probe cleaning

Why?

Impurities on the probe surface



Typical double probe characteristic

May contaminate the plasma

May provoke secondary emission from the probe surface

To see this effect for clean probe $T^e$ should be >15 eV *

Distortion of the probe I-V

Chemically active plasma

solution

solution

Electron current cleaning

Ion bombardment

For thin film removing

Typical cleaning voltage ± (0…100 V) and current -1..2mA

Direct or indirect** continuous heating***

*Amemia and Fuchs Jpn. Appl. Phys 30, 3531 (1991)

** J. Kalčík, *Czech. J. Phys.* **45**(1995)241

***M. Niionomi, K. Yanagihara, *ACS Symp.* 1088, 7.(1979)

# Langmuir probe and RF discharge

13.56 MHz RF discharge

Argon: $n_i \approx n_e = 10^{10}$-$10^{11}$ cm$^{-3}$

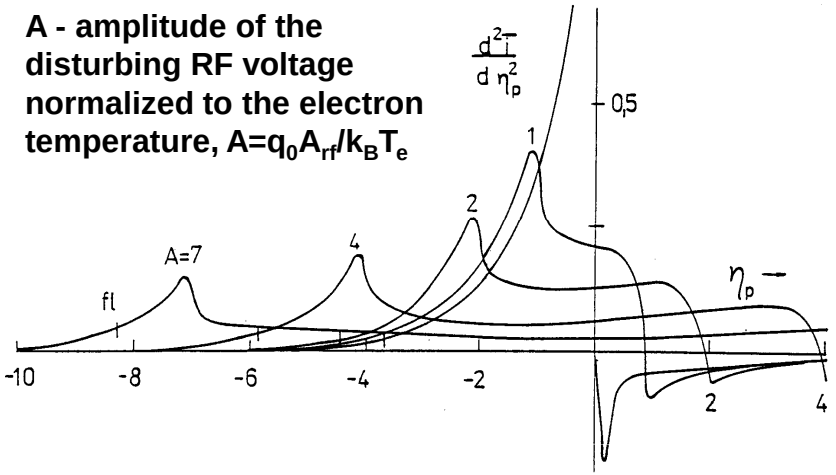
Electrons follow changes of RF Field $\omega_{pe} > 1$ GHz

Ions (no response) $\omega_{pi} \approx$ 3-10 MHz

The increase of the electron probe current in RF frequency range degrades the probe characteristic

Probe measurements ⇨ useless.

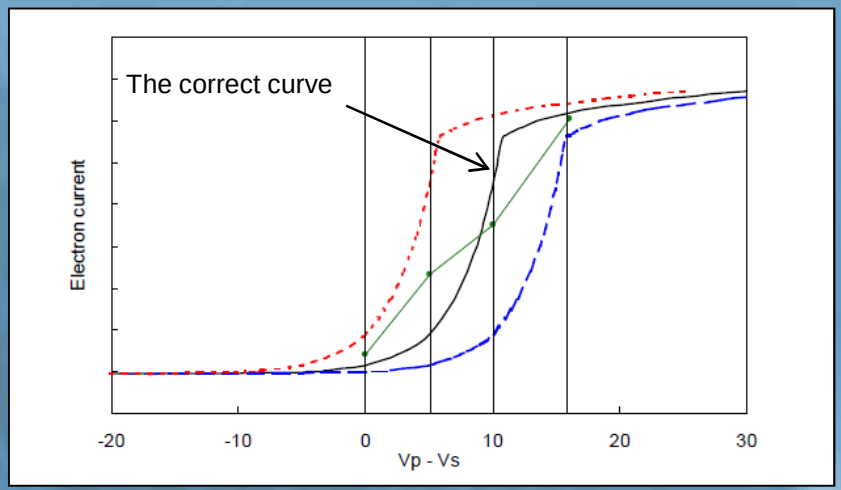
**A - amplitude of the disturbing RF voltage normalized to the electron temperature, $A=q_0A_{rf}/k_BT_e$**




Distortion of the I" caused by RF voltage (sinus shaped)

*S. Klagge, M. Maas, Beitr. Plasmaphys. 23(1983)487*

The dashed ones are displaced by ±5V, representing changes in Vs. At the vertical lines, the average Ie between the displaced curves is shown by the dot. The line through the dots is the time-averaged I - V curve that would be observed, differing greatly from the correct curve.



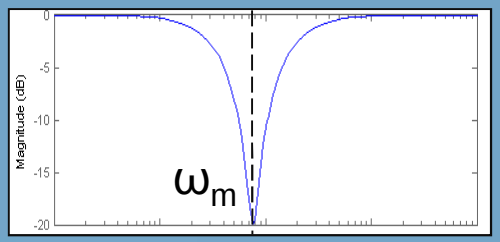

F.F. Chen Langmuir probe diagnostics, IEEE-ICOPS meeting, Jeju, Koreja, 2003

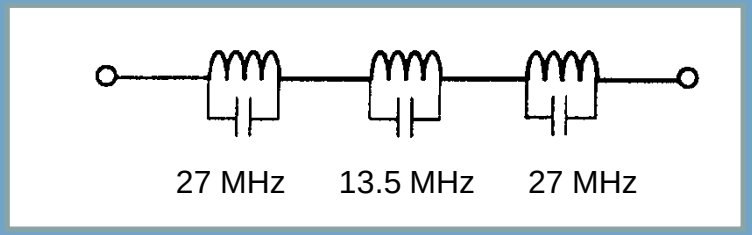
# Compensation methods of the RF probe

## Passive compensation

**Band stop filter**

$$\omega_m = \sqrt{\left(\frac{1}{2R_0C}\right)^2 + \frac{1}{LC}}.$$

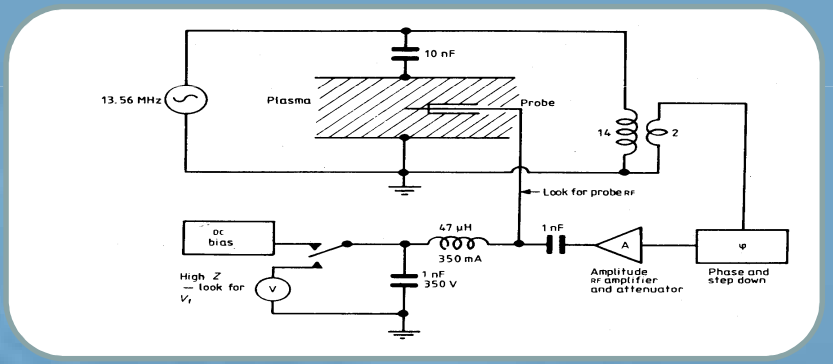



*V.A. Godyak, R.B. Piejak, B.M. Alexandrovich, Plasma Sources Sci. Technol. 1, 36 (1992)*

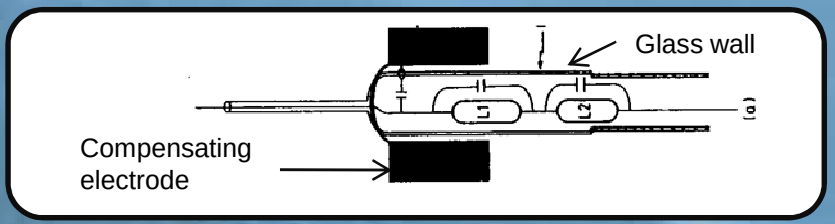
## Active compensation

N.St.J. Braithwaite, N.M.P. Benjamin,
J.E. Allen, J.Phys. E Sci. Instr. 20(1987)1046




## Combined compensation

P.A. Chatterton, et al, Vacuum 42(1991)489
M Aflori and J L Sullivan Romanian Reports in Physics, Vol. 57, No. 1, P. 71–77, 2005
U. Flender et alPlasma Sources Sci. Technol. 5(1996)61



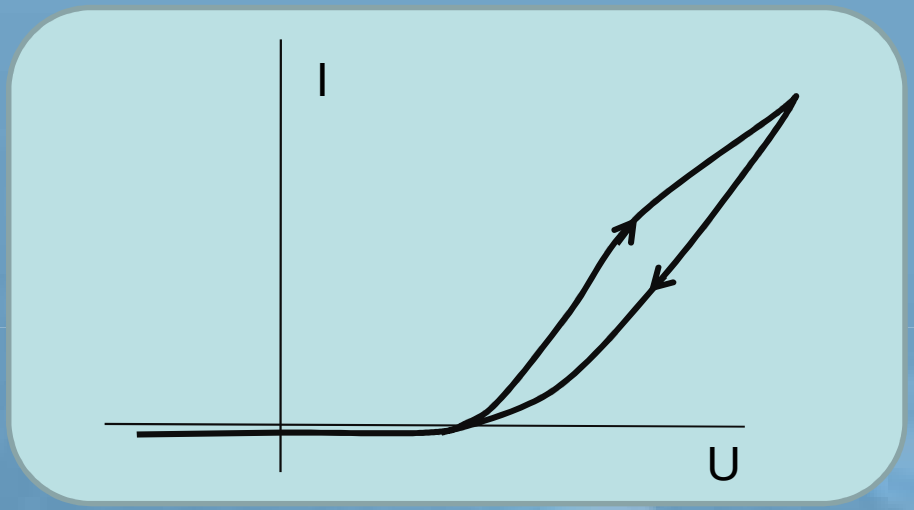

Comparison:
Experimental set-up:

B.M. Annaratone, N.St.J. Braithwaite, Meas. Sci. Technol. 2, 795 (1991)

H. Sabadil, S. Klagge, M. Kammayer, Plasma Chemistry and Plasma Processing 8,425 (1988)

# Probes in the chemically active plasma

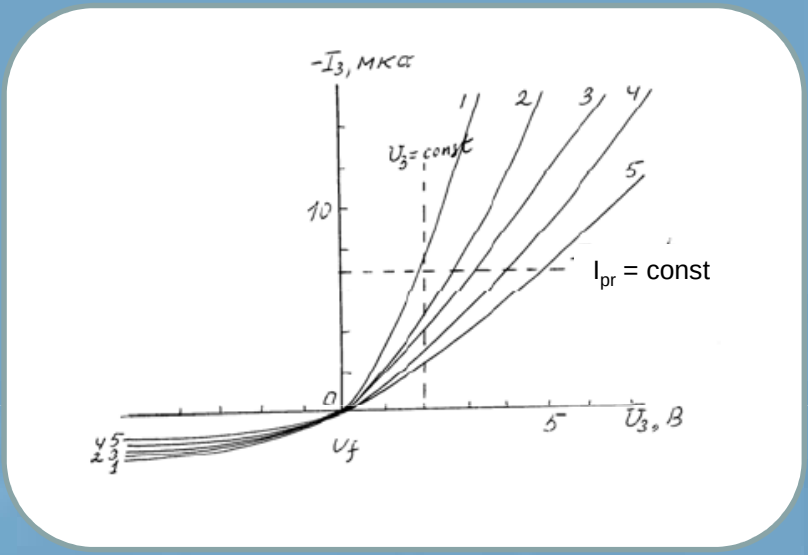
Distortion of the probe characteristic due to the non-conducting layer on the probe surface and/or on the reference electrode.




**Solution:**

| |
|---|
| **Heating, ion bombardment** |
| **Continuous cleaning**<br>Warm probe<br>Emissive probe |
| **Systems without a reference electrode**<br>Double probe<br>Triple probe<br>Another special probes |

**Thin Film Deposition**

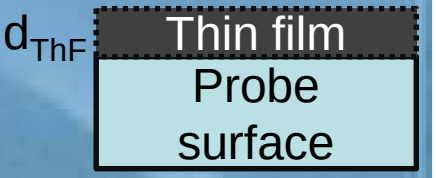



$I_{pr}$ = const

$\Delta U_{pr} = I_{pr} R_{ThF}$

$d_{ThF} = R_{ThF}/\rho_{ThF}$

$\Delta d_{min} \approx 2\text{-}5nm$

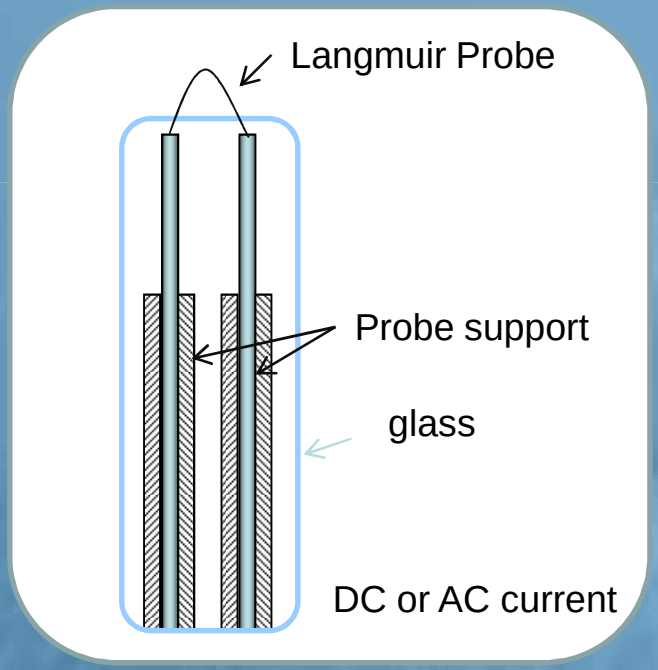



Овсянников А. А., Энгельшт В. А., Лебедев Ю. А. и др. Диагностика низкотемпературной плазмы.Новосибирск: Наука, 1994.

# Heated probes

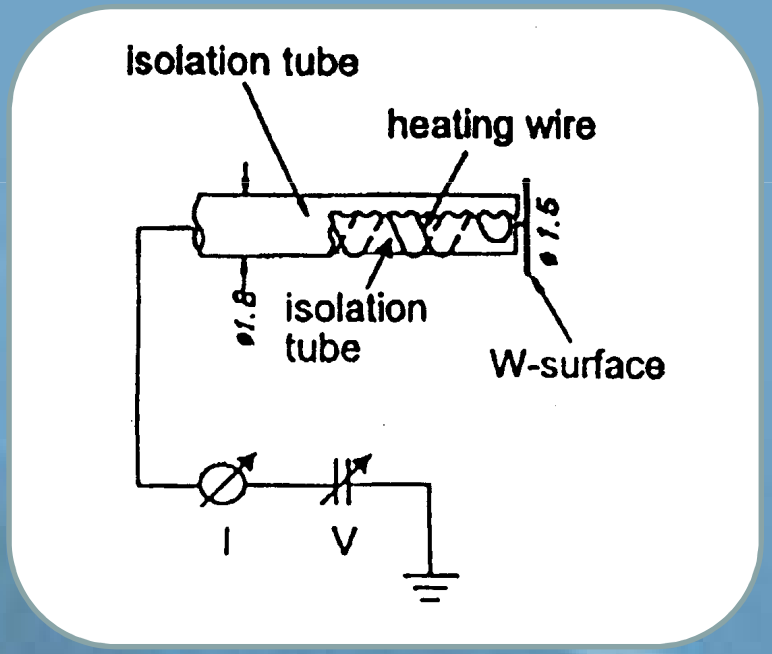
## Directly heated probes

warm or emissive




J. Kalčík, Czech. J. Phys. 45(1995)241

## Indirectly heated probes

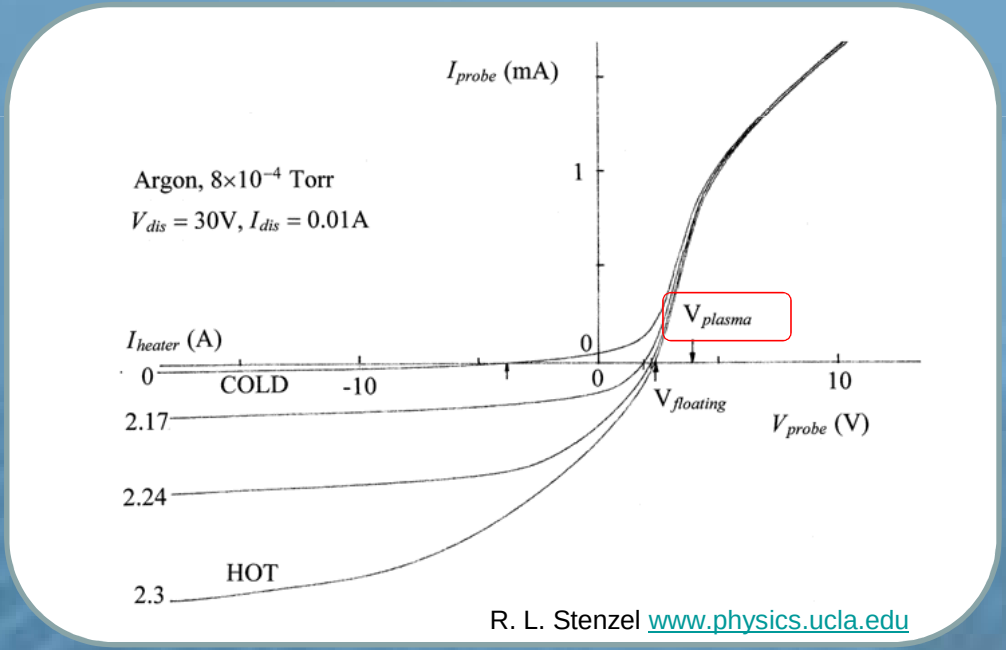



C. Winkleret al Proc. 12th SAPP, Liptovský Ján (Slovakia), February, 9-13, 1999, J.D. Skalný, M. Černák, Eds., p 121

# Emissive probes

Direct measurement of the **plasma potential**.
Electric field measurements.

Do not give useful information about plasma density and $T_e$




R. L. Stenzel www.physics.ucla.edu

**The floating potential**, defined as the voltage at zero probe current, **shifts toward the plasma potential**. The larger the emission current, the closer the emissive probe floats to the plasma potential.

$V_{pl}$ can be approximately obtained from the open-loop voltage of an emissive probe.

$E=-\text{grad}V_{plasma}$

$$V_{pl} = V_{fl} + T_e \left( \frac{I_{-sat}}{I_{+sat} + I_{em}} \right)$$

$$I_{em} \Rightarrow \infty \quad \Rightarrow \quad V_{fl} \approx V_{pl}$$

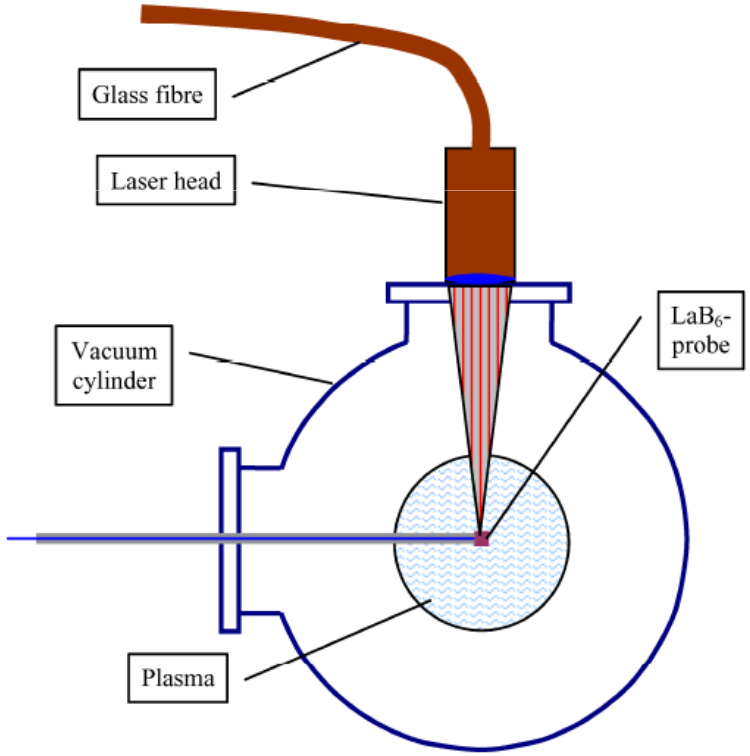
E.Y. Wang, N. Hershkowitz, T. Intrator, C. Forest, Rev. Sci. Instr. 57(1986)2425

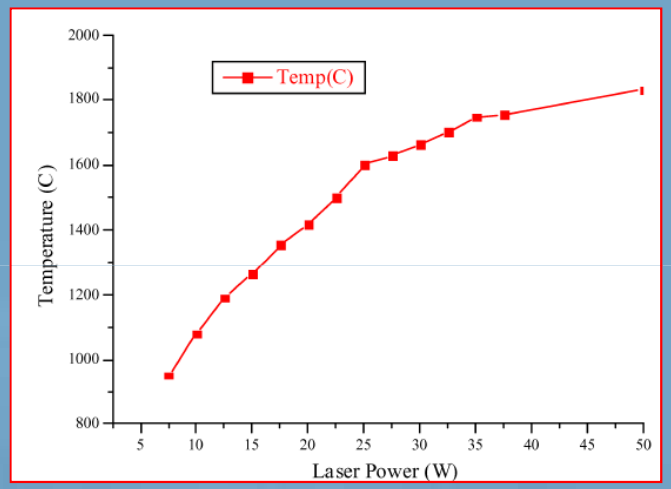
# Emissive probes

Disadvantages of the **loop-probe** design are the **limited life time** of the tungsten wire, the **low electron emissivity** of tungsten and the voltage drop across the wire.

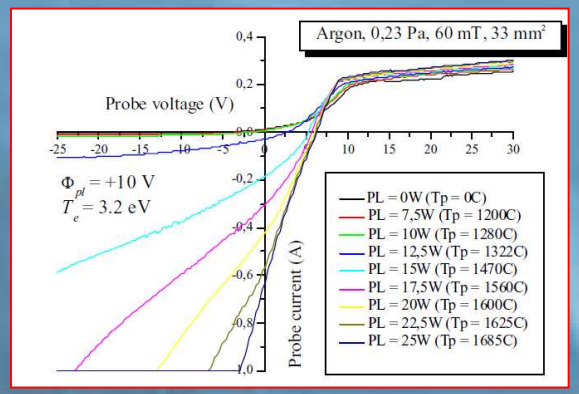
## A Laser - Heated Emissive Probe for Fusion Applications

**Emission current up to 20A**



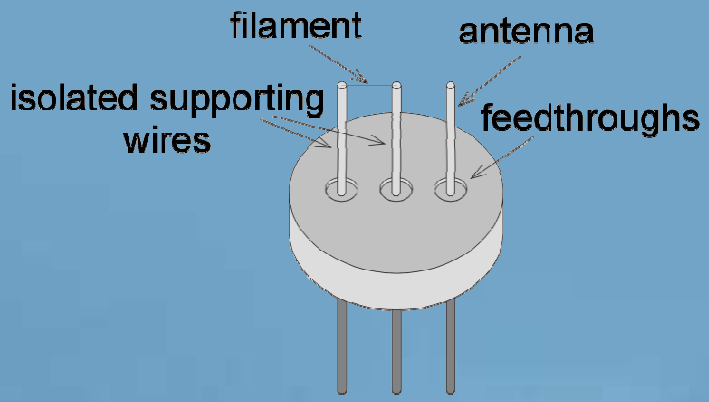




Increase of the temperature of the probe piece with the laser power




R. Madani, C. Ionita, R. Schrittwieser, G. Amarandei, P. Balan, T. Klinger 31st EPS Conference on Plasma Phys. London, 28 June - 2 July ECA Vol.28G, P-5.127 (2004)

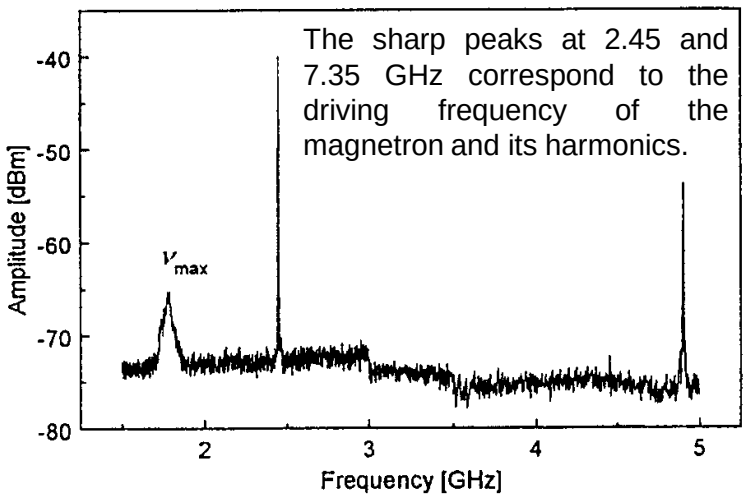
# Plasma oscillation probe (POP)



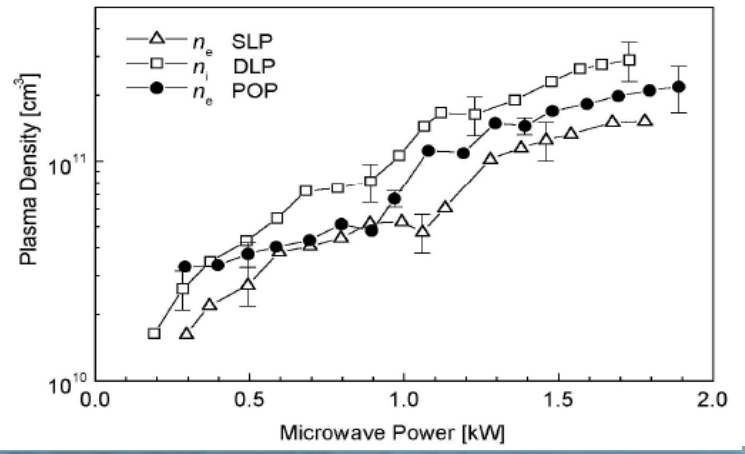

$$\nu_{\max} = \omega_{pe}\left[1 - \frac{1}{2}\left(\frac{1}{2}\eta\right)^{1/3} + \ldots\right]$$

$$\eta = n_{beam} / n_e << 1 \qquad \varepsilon_{beam} >> \varepsilon_{pe}$$

$$\nu_{\max} \approx \omega_{pe} = \sqrt{\frac{q^2 n_e}{\varepsilon_0 m_e}}$$

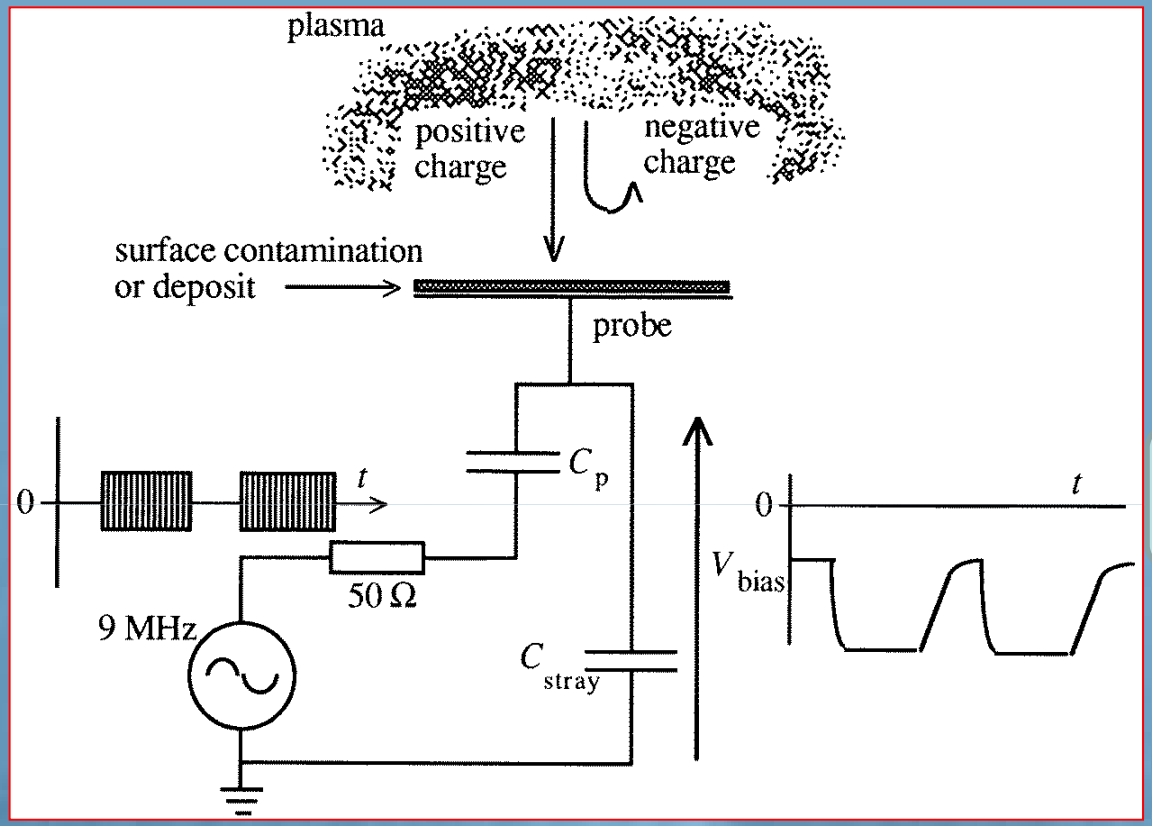


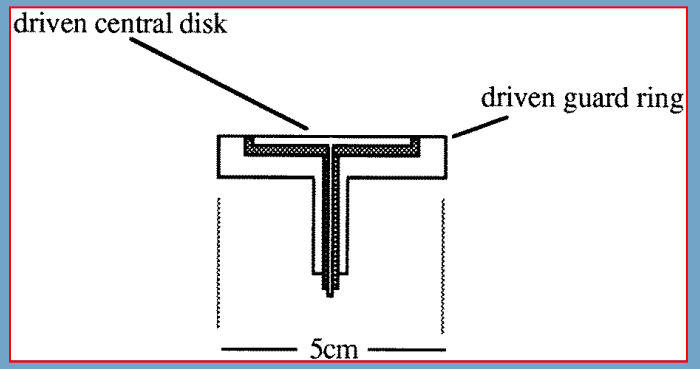

Frequency spectrum in an underdense argon microwave plasma at 3 Pa and 800 W input power




*A. Brockhaus, A. Schwabedissen, Ch. Soll, J. Engemann, Proc. Frontiers in Low Temperature Plasma Diagnostic III, Saillon (Switzerland), 15-19 February 1999, p. 89*

# RF probe. Ion flux measurements




$$\frac{dV_{bias}}{dt} = eA\left(\Gamma_i + \Gamma_e(V_{bias})\right) / C_p$$

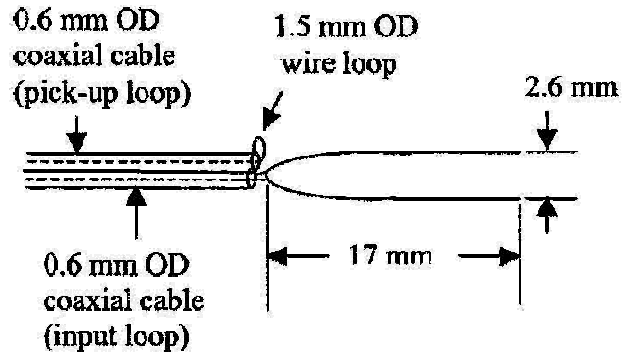
A – probe area

The RF signal to the probe electrode is periodically chopped and the change in potential across the capacitor–generator combination is monitored. If the magnitude of he RF bias is chosen large enough for $\Gamma_e(V_{bias})$ to be zero after the RF is chopped off, the ion flux is easily determined from the initial rate of discharge.

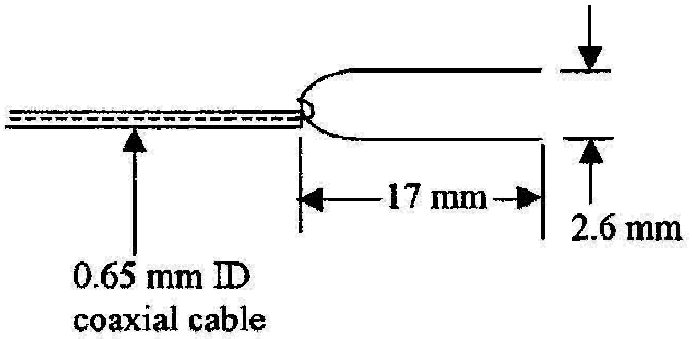
*N. St.J. Braithwaite, J.P. Booth, G. Cunge, Plasma Sources Sci. Technol. 5(1996)677*

# Hairpin resonance probe

The technique depends only on the local electron density and is independent of the ion fluxes received by the probe.




A schematic diagram of a transmission type hairpin resonator.




A schematic diagram of a reflection type hairpin resonator.

The principle is based on measuring the plasma dielectric constant, ε, using a U - shaped resonant structure. The resonant structure when placed in plasma, exhibits a shift in the characteristic resonance as in vacuum. From the measured frequency shifts, the electron density is directly determined.

$$f_r = \frac{c}{4L\sqrt{\varepsilon}}$$

$f_0$ the resonant frequency in vacuum

L – resonator length,

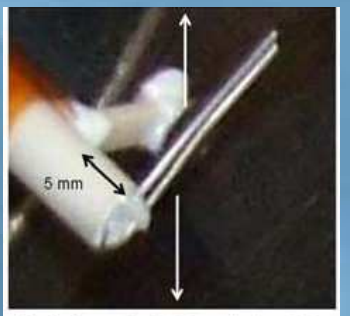
ε relative permittivity

$f$ in GHz

$$\varepsilon = 1 - \frac{f_p^{\,2}}{f^{\,2}}$$

$$f_p = \left(\frac{q_0 n}{\pi m}\right)^{1/2}$$

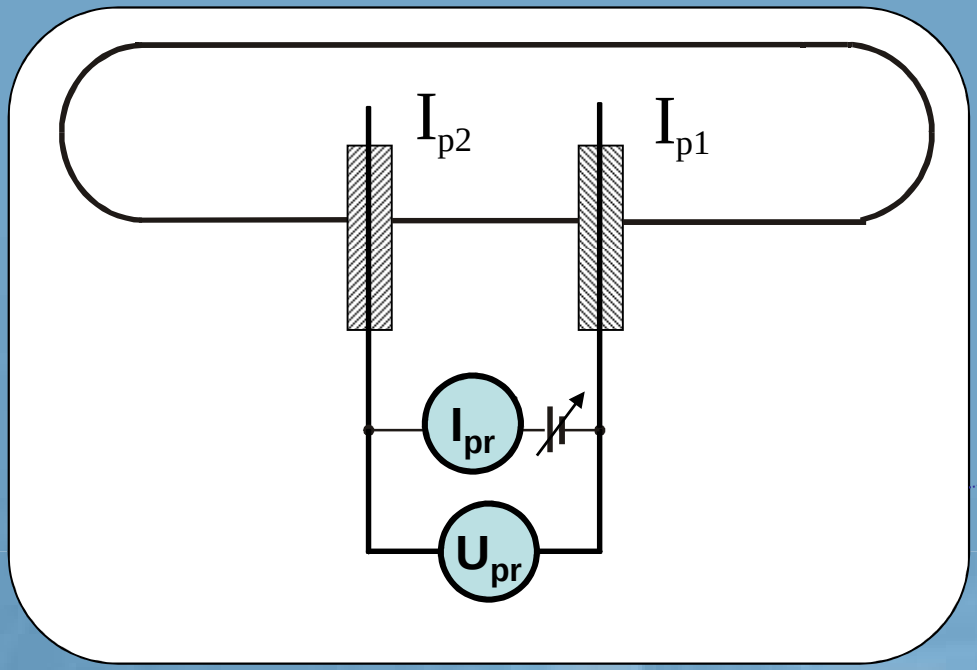
$$n = \frac{f_r^{\,2} - f_0^{\,2}}{0.81} \times 10^{10} \text{ cm}^{-3}$$



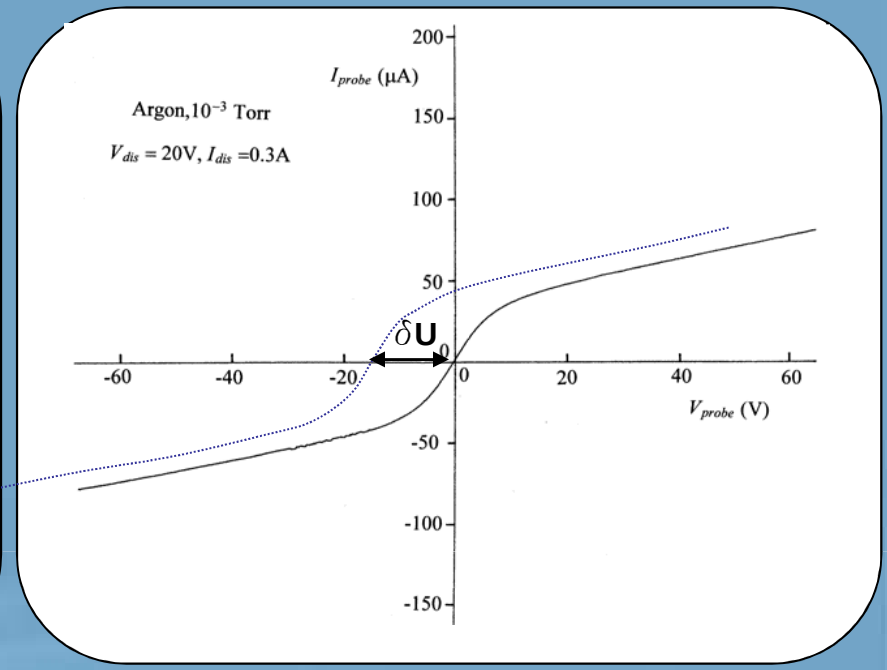

Hairpin probe (1.5 cm in length)

R. L. Stenzel, Rev. Sci. Instrum. 47 (1976) 603
Piejak R.B., et al., J. Appl. Phys., Vol. 95 (2004) 3785
S. K. Karkari et al., 28th ICPIG, July 15-20, 2007,1624

# Double probe




Typical double-probe circuit




Typical characteristic of double probe

Electric field in a discharge plasma

Electron temperature

Plasma density

No large reference electrode

~~EEDF~~ ~~Plasma potential~~

solution

Triple probe

Double probe: E.O. Johnson, L. Malter, Phys. Rev. 80(1950)58; J.S. Chang, J.G. Laframboise, *Phys. Fluids* **19**(1976)25

Triple probe: S. Chen, T. Sekiguchi, J. Appl. Phys. 36(1965)2363; J.S. Chang, et al .J.Phys.D: Appl. Phys. 13(1980)1083

# Summary

**Advantages of the probe diagnostics method**

- Comparatively simple biasing electronics and experimental set-up
- Plasma parameters: plasma density ($n_e$, $n_i$), temperature ($T_e$), EEDF,ion flux, plasma potential
- Time and space-resolved measurements
- Good spatial (~$\lambda_D$) and temporal (~$2\pi/\omega_i$, ) resolution

**Disadvantages**

- The presence of the probe in the plasma can initiate instabilities in the plasma.
- The plasma surrounding the probe is disturbed by the drain of the charge carriers to the probe.
- The application of probe method is more difficult or impossible in plasma containing fluctuations, oscillations and waves.
- The heavy particles can cause the creation/destruction of thin films on the probe surface during measurements which change the properties of the probe surface. Thus the probe data being difficult or impossible to interpret.
- Difficult to use in high temperature plasma devices (TOKAMAK etc)

M Tichy et al., Plasma Sources Sci. Technol. 18 (2009) 014009

# Thank you for attention!